



نايف الراجحي الاستثمارية
NAIF ALRAJHI INVESTMENT

**REQUEST FOR INFORMATION (RFI)
FOR
THE IMPLEMENTATION OF SMART CITY
SOLUTIONS
AT**

حديقة الملك سلمان
King Salman Park



DISTRICT II

RFI No.: XXXXX

Date: 19th October, 2025

Disclaimer:

The issue of this RFI document does not imply that Naif Alrajhi Investment (NRI) is bound to select a bidder or to appoint the selected bidder, for implementation and NRI reserves the right to reject all or any of the bidders or bids without stating any reason. The bidder shall bear all its costs associated with or relating to the preparation and submission of its bid including but not limited to preparation, copying, postage, delivery fees, expenses associated with any demonstrations or presentations which may be required by NRI or any other costs incurred in connection with or relating to its bid. All such costs and expenses will remain with the bidder and NRI shall not be liable in any manner whatsoever for the same or for any other costs or other expenses incurred by a bidder in preparation for submission of the bid.

Important Guidelines:

- Interested parties must demonstrate the following:
- Provide original brochures or Internet technical data sheets from a known source in English language.
- All offered Hardware and Software must be latest production.
- All offered solutions and devices must comply with the relevant standards of Kingdom of Saudi Arabia.
- The offered items (Software & Hardware) and their options must be priced individually to the level of the unit.
- Vendors and suppliers participating in the proposal either directly or through a representative should have a good representation in Saudi Arabia. Full details about the vendor- representative relationship should be submitted if any.
- All Software components must support English and Arabic.
- The bidder must provide the full maintenance support for his offered system in Saudi Arabia including maintenance procedures and capabilities.
- Bidder must provide:
 - Five years warranty for all items starting from the date of final handover.
 - Five years of full system support including Software and Hardware starting from the date of final handover.
 - Five years of software development and integrations support.
- The proposal from interested parties must demonstrate the ability to guarantee the performance of all system components concerning the immediate requirements and system growth in the future.
- MSI shall respond using the same section numbering and structure as this RFI.
- Responses must be comprehensive, technically validated, and supported by relevant datasheets, architecture diagrams, and compliance certificates.
- MSI must provide the list of all possible licensed features of each offered Software Solution as per below table:

S. No.	Part No.	Description	Type (Perpetual / Subscription)	Based on (System Wide / No. of Sensors No. of Servers / No. of instances / No. of Cores etc.)

Table of Contents

1. Introduction:	5
1.1. King Salman Park:	5
1.2. NRI District II:	5
1.3. RFI Objective:	5
2. Warranty, Maintenance and Support:	7
2.1. Applications Support and Maintenance	7
2.2. Warranty	7
3. Functional Requirements:	9
3.1. Use Case 1 – Digital Payment Platform	9
3.2. Use Case 2 – Computer Aided Facilities Management System (CAFM)	11
3.3. Use Case 3 – Integrated Safety and Security Platform (CCTV, ACS, SMS)	14
3.4. Use Case 4 – Intelligent Lamp Posts and Smart Poles	18
3.5. Use Case 5 – Mobile App (Citizen Engagement Experience App)	22
3.6. Use Case 6 – Smart Waste Management	26
3.7. Use Case 7 – Electric Vehicles Charging Stations & Points (EVCS)	29
3.8. Use Case 8 – Security Command & Control Center / Operations Control Room	32
3.9. Use Case 9 – IoT Platform (Central IoT Management)	36
3.10. Use Case 10 – Integrated Corporate / Operations Billing System	39
3.11. Use Case 11 – Building Information Modelling (BIM)	42
3.12. Use Case 12 – Electronic Document Management System (EDMS)	45
3.13. Use Case 13 – Public Address & Voice Alarm Systems (PAVA)	48
3.14. Use Case 14 – Core, Consolidated Smart Cities Platform	51
3.15. Use Case 15 – Enterprise Resource Planning (ERP)	55
3.16. Use Case 16 – Smart Irrigation Systems & Water Management	58
3.17. Use Case 17 – Building Access Control System (BACS)	61
3.18. Use Case 18 – Addressable Fire Alarm System (ICT Scope)	64
3.19. Use Case 19 – Building Management System (BMS)	67
3.20. Use Case 20 – Public Help Desk & Call Centre System Solution	70
3.21. Use Case 21 – Energy Management System (EMS)	73
3.22. Use Case 22 – Security Video System & Control System (CCTV Surveillance – ICT Scope)	76

1. Introduction:

1.1. King Salman Park:

King Salman Park, the largest urban park in the world, sits at the heart of Riyadh and offers a wide range of activities, from arts and culture experiences to sports, innovation, and entertainment features.

Launched in 2019 by King Salman bin Salman bin Abdulaziz, King Salman Park sits at the heart of Riyadh. As the largest urban park in the world, it will offer a wide range of activities and experiences for visitors of all ages.

The Park's vast open green spaces make it a desired venue for hosting sports and cultural events, and artistic and recreational activities, while its relaxing environment enables visitors to experience nature. The Park also includes an Islamic style garden, vertical gardens, a maze garden, and a bird and butterfly sanctuary.

In addition to its cultural and recreational offerings, King Salman Park's design has sustainability and environmental protection in mind, with goals to increase the vegetation in the region and raise the rate of per capita green spaces.

1.2. NRI District II:

The Project consists of 31 mixed use buildings including residential zones, mixed-use areas for retail and offices, an educational institution, community centers, and hotel with an area approx. 290,000 m². Head Quarter of King Salam Park will also be located here.

The district aims to provide lifestyle community living centered around enhancing the quality of life for residents by using technology, data, and sustainable practices.

1.3. RFI Objective:

NRI intends to select a master system integrator (MSI) / consortium to participate in the design, development, implementation, maintenance and integration of the smart solutions starting from the date of awarding the contract and for a period of five years after the go live date. This RFI contains the scope of work that will be assigned to the MSI as part of this project.

The MSI will develop an integrated smart city solution for NRI with the focus on the below tracks:

- Main Smart City Platform
- Digital Payment Platform
- Computer Aided Facilities Management System (CAFM)
- Integrated Safety and Security Platform (CCTV, ACS, SMS)
- Intelligent Lamp Posts and Smart Poles

- Mobile App (Citizen Engagement Experience App)
- Smart Waste Management
- Electric Vehicles Charging Stations & Points (EVCS)
- Security Command & Control Center / Operations Control Room
- IoT Platform (Central IoT Management)
- Integrated Corporate / Operations Billing System
- Building Information Modelling (BIM)
- Electronic Document Management System (EDMS)
- Public Address & Voice Alarm Systems (PAVA)
- Enterprise Resource Planning (ERP)
- Smart Irrigation Systems & Water Management
- Building Access Control System (BACS)
- Addressable Fire Alarm System (ICT Scope)
- Building Management System (BMS)
- Public Help Desk & Call Centre System Solution
- Energy Management System (EMS)
- Security Video System & Control System (CCTV Surveillance – ICT Scope)

The scope of work for the MSI is to implement the Smart City System. The scope includes software/solution development and implementation, Information Technology (IT) and maintenance of the Smart City system including the citizen engagement portal and mobile application. The implementation period for this RFI is 2 years. The maintenance phase will be for a period of five years after Go-Live. Post completion of the 5-year period, the contract can be extended, at discretion of NRI. MSI needs to design, implement and maintain the Smart City System project on turnkey basis. MSI needs to do the appropriate solution design and sizing for the project as per the scope of work of the RFI. In case MSI has not considered any component/service, which is necessary for the project requirement, the same needs to be brought by the MSI at no additional cost to NRI.

2. Warranty, Maintenance and Support:

The Smart City Solution include innovative and integrated use cases served from variety of systems. The most important aspect is to provide an integrated solution that can integrate and collect data from multiple sources helping NRI to make better and more efficient decisions. NRI should be able to view a single instance of truth from the same data pool using a single interface.

2.1. Applications Support and Maintenance

Application support includes, but not limited to, production monitoring, troubleshooting and addressing the functionality, availability and performance issues, implementing the system change requests etc. The MSI shall keep the application software in good working order; perform changes and upgrades to applications as requested by the NRI team. Key activities to be performed by MSI in the application support phase are as follows:

- The MSI should have a sufficient team on site to provide application and support as needed. MSI need to provide the proposed team structure including number of team members proposed to be deployed along with roles and skills of each such member.
- The MSI should ensure that its development team works closely with NRI development team while providing knowledge transfer plan and trainings that will enable NRI team to continue the operation and customizations in the future.
- MSI shall address all the errors/bugs/gaps in the functionality in the solution implemented by the MSI at no additional cost during the Support and Maintenance phase
- Any changes/upgrades to the software performed during the support phase shall subject to the comprehensive and integrated testing by the MSI to ensure that the changes implemented in the system meets the specified requirements and doesn't impact any other function of the system. Release management for application software will also require NRI approval.
- Monthly report on problem identified and resolved would be submitted to NRI team along with the recommended resolution.
- The bidder must clarify in his proposal the support lifecycle. This should also include the product roadmap for any of the proposed applications.

2.2. Warranty

- MSI shall provide comprehensive and on-site warranty for 5 years from the date of Go-Live for all equipment.
- MSI shall provide the comprehensive & onsite manufacturer's warranty in respect of proper design, quality and workmanship of all hardware, equipment, accessories etc. covered by the RFI. MSI must warrant all hardware, equipment, accessories, spare parts, software etc. procured and implemented as per this RFI against any manufacturing defects during the warranty period.
- MSI shall provide the performance warranty in respect of performance of the installed hardware and software to meet the performance requirements and service levels in the RFI.
- MSI is responsible for sizing and procuring the necessary hardware and software licenses as per the performance requirements provided in the RFI. During the warranty period MSI shall replace, augment, or procure higher-level new equipment or additional licenses/hardware at no additional

cost to the NRI in case the procured hardware or software is not enough or is undersized to meet the service levels and the project requirements.

- The MSI shall carry out Preventive Maintenance (PM) of all hardware and software, and should maintain proper records at each site for such PM. The PM should be carried out at least once in six months as per checklist and for components agreed with NRI.
- The MSI shall carry out corrective maintenance for maintenance/troubleshooting of supplied hardware/software. The MSI shall also maintain complete documentation of problems, isolation, and cause and rectification procedures for building knowledge base for the known problems in centralized repository, accessible to NRI team as well.
- MSI shall have to stock and provide adequate onsite and offsite spare parts and spare component to ensure that the uptime commitment as per SLA is met.

3. Functional Requirements:

3.1. Use Case 1 – Digital Payment Platform

1. Background and Objective

As part of the Smart Community initiative, the Digital Payment Platform aims to provide a unified and secure method for digital transactions within the NRI ecosystem. The platform will support various use cases such as ticketing, access to park facilities, parking fees, and other visitor-related services.

The objective of this RFI is to gather comprehensive technical and operational information from the MSI regarding the proposed solution architecture, system capabilities, integration approach, security compliance, and operational requirements to ensure a seamless implementation aligned with the overall Smart Community framework.

2. Scope of Work

The scope of this Use Case includes, but is not limited to, the design, supply, implementation, configuration, and commissioning of a Digital Payment Platform that enables end-to-end payment transactions for residents, visitors and operators.

The system should:

- Support multiple payment methods (credit/debit cards, mobile wallets, NFC, QR code, wearables).
- Enable seamless transactions across all park facilities (retail, events, parking, etc.).
- Provide a centralized management interface for monitoring transactions, revenues, and analytics.
- Integrate with other Smart Community systems such as the Electronic Ticketing System, ERP, Mobile App (Visitor Experience), and IoT Platform.
- Ensure compliance with global and local standards for payment security, privacy, and data protection (e.g., PCI DSS, ISO 27001, GDPR-equivalent).
- Support multi-language interfaces (Arabic/English) for both end-users and administrative portals.

3. Technical Requirements

3.1 System Architecture

- Describe the proposed end-to-end architecture, including system components, communication flow, data exchange layers, and security zones.
- Identify whether the platform will be cloud-based, on-premises, or hybrid, and specify hosting requirements.
- Provide redundancy, scalability, and high-availability mechanisms.

3.2 Functional Capabilities

- Multi-channel payment processing (POS, kiosk, web, mobile app, wearable devices).
- Integration with digital wallets and third-party payment gateways.

- Transaction management dashboard and analytics reporting.
- Refund, loyalty, and promotional features.
- Role-based access control and audit trail capabilities.

3.3 Integration and Interoperability

Describe how the platform will integrate with:

- Electronic Ticketing System & Wearables
- Mobile App (Visitor Experience)
- ERP / Billing System
- IoT Platform (Central Data Hub)
- Define API standards, data formats, and authentication methods.
- Provide details on how real-time transaction data will be shared with other systems.

3.4 Cybersecurity and Data Privacy

- Explain compliance with PCI DSS and other payment industry security standards.
- Describe encryption methods for data in transit and at rest.
- Define authentication and authorization mechanisms for users and administrators.
- Clarify data retention policies, anonymization practices, and compliance with privacy regulations.

3.5 User Interface and Experience

- Describe the design approach for both front-end and back-office interfaces.
- Ensure responsive design for various devices (kiosks, mobile, tablets).
- Provide support for multi-language and accessibility features.

4. Operational and Maintenance Requirements

- Outline the proposed maintenance model, including SLAs, incident response, and performance monitoring.
- Define procedures for system updates, security patches, and version control.
- Specify any required licenses, warranties, and long-term support commitments.
- Describe the training plan for system administrators and operators.

5. Deliverables

MSI is expected to provide the following deliverables as part of this RFI response:

- Detailed system architecture and design proposal.
- Product datasheets and technical specifications.
- Reference projects and implementation case studies.
- Integration methodology and data flow diagrams.
- Compliance certificates (PCI DSS, ISO 27001, etc.).
- Operations and maintenance plan.

3.2. Use Case 2 – Computer Aided Facilities Management System (CAFM)

1. Background and Objective

The Computer Aided Facilities Management System (CAFM) is one of the core operational systems within the Smart Community framework, intended to support efficient maintenance management, asset tracking, work order processing, and resource planning.

This system shall enable the facility management teams to plan, execute, and monitor all maintenance and operations activities; both reactive and preventive; across the entire community infrastructure.

It will also serve as the digital backbone connecting maintenance operations with other Smart Systems such as IoT, BMS, EMS, ERP, and the C4 Command Center.

The objective of this RFI is to obtain detailed technical and functional information from the MSI regarding the proposed CAFM solution, ensuring its scalability, interoperability, and compliance with international facility management standards (ISO 41001).

2. Scope of Work

MSI shall design, supply, configure, and deploy a Computer Aided Facilities Management System (CAFM) that includes:

- Asset registration and lifecycle management.
- Preventive, corrective, and predictive maintenance management.
- Work order automation and field service tracking.
- Integration with IoT sensors for condition-based maintenance.
- Integration with mobile devices for technician access.
- Reporting, analytics, and performance dashboards.
- Integration with other systems: ERP, BMS, and Energy Management (EMS).
- Centralized access through web and mobile interfaces.

3. Technical Requirements

3.1 System Architecture

Provide detailed system architecture including servers, applications, databases, and integration layers. The solution must support:

- On-Premises, Cloud-based or hybrid deployment.
- Multi-tenant and multi-site structure.
- API-based integration with external systems.
- Data synchronization in real time with IoT and ERP systems.
- Ensure modular design for scalability and future expansion.
- Comply with ICT infrastructure and cybersecurity standards.

3.2 Functional Capabilities

Asset Management:

- Registration, tagging, and classification of all community assets.
- Asset hierarchy management (location, type, function).
- Condition monitoring and lifecycle tracking.

Maintenance Management:

- Preventive, corrective, and predictive maintenance scheduling.
- Automated work order generation based on IoT data or thresholds.
- Technician mobile app for real-time updates and status reporting.

Workforce and Resource Management:

- Resource allocation and shift planning.
- Contractor and vendor performance tracking.
- Integration with HR or ERP for labor cost calculation.

Inventory and Procurement:

- Spare parts inventory and procurement management.
- Integration with ERP for stock, purchase orders, and invoices.

Analytics and Reporting:

- KPI dashboards (MTBF, MTTR, SLA compliance).
- Energy, cost, and asset performance analytics.
- Customizable reports with data export options.

3.3 Integration and Interoperability

The CAFM System shall integrate with:

- IoT Platform (Use Case 9): For real-time data and predictive maintenance triggers.
- BMS (Use Case 19): For automated work orders based on system alarms.
- EMS (Use Case 21): For energy performance correlation.
- ERP (Use Case 15): For financial, procurement, and HR data exchange.
- C4 Center (Use Case 8): For centralized incident visualization and command.
- Smart Cities Platform (Use Case 14): For unified dashboards and data analytics.
- Integration shall be achieved using open APIs (RESTful), middleware, or message queues.
- All systems must comply with interoperability and data-sharing standards defined by the Smart Community's ICT architecture.

3.4 Cybersecurity and Data Privacy

- Comply with ISO 27001 and NIST Cybersecurity Framework.

- Secure data encryption (AES-256 at rest, TLS 1.3 in transit).
- Role-based access control (RBAC) and multi-factor authentication.
- Audit trail for all user actions and configuration changes.
- Regular penetration testing and vulnerability assessment.
- Backup, disaster recovery, and data retention policies.

3.5 User Interface and Experience

- Web-based interface accessible via desktop and mobile.
- Multilingual support (Arabic/English).
- GIS integration to visualize asset locations and work orders.
- Intuitive dashboards for technicians, supervisors, and management.
- Notification and alert system (email, SMS, mobile push).
- Role-based customization of dashboards and reports.

4. Operational and Maintenance Requirements

- Define SLAs for system uptime ($\geq 99.9\%$) and response time.
- Provide preventive and corrective maintenance plans.
- Include system documentation, user manuals, and as-built configurations.
- Provide user and administrator training sessions.
- Implement version management and software updates.
- Ensure warranty and support for at least 5 years.
- Regular performance optimization and system audits.

5. Deliverables

- MSI shall include the following as part of the RFI response:
- System architecture and integration diagrams.
- Product datasheets and software module descriptions.
- Integration plan with IoT, ERP, and BMS.
- Cybersecurity and compliance documentation.
- Maintenance and training plan.
- Example dashboards and analytics reports.



3.3. Use Case 3 – Integrated Safety and Security Platform (CCTV, ACS, SMS)

1. Background and Objective

As part of the Smart Community development, the Integrated Safety and Security Platform (CCTV, ACS, SMS) aims to provide a unified, intelligent, and resilient security infrastructure ensuring the protection of people, assets, and facilities across the entire community.

This platform will consolidate the Closed-Circuit TV (CCTV) system, the Access Control System (ACS), and the Security Management System (SMS) into a single, interoperable ecosystem managed through a centralized interface.

The objective of this RFI is to collect comprehensive technical, operational, and integration details from the MSI regarding the proposed safety and security architecture. The goal is to achieve real-time situational awareness, seamless event correlation, and centralized command and control capabilities, ensuring a proactive approach to community safety and emergency response.

2. Scope of Work

MSI shall design, supply, install, configure, and commission an Integrated Safety and Security Platform encompassing the following systems and subsystems:

- CCTV Surveillance System: For real-time video monitoring, analytics, and recording.
- Access Control System (ACS): For managing entry and exit across facilities using credentials such as cards, mobile access, or biometrics.
- Security Management System (SMS): To act as the central platform integrating all security subsystems, handling alarms, incidents, and event management.
- Integration with Fire Alarm System, Building Management System (BMS), and Security Control Center (C4).
- Compliance with international and local security standards such as ONVIF, UL 294, ISO 27001, and relevant civil defense regulations.
- The platform must enable unified management of all safety and security devices and ensure interoperability between all subsystems to support operational efficiency, threat detection, and automated incident response.

3. Technical Requirements

3.1 System Architecture

- Provide a detailed overview of the proposed integrated security architecture, showing all system layers (field devices, communication, application, and command center).
- Indicate the topology (distributed or centralized), and highlight redundancy, failover, and disaster recovery design.
- Define connectivity among subsystems (CCTV, ACS, SMS) using secure protocols.
- The platform should support scalability for future system expansion and integration of additional sensors or applications.

- Clearly define how each subsystem communicates with the central Security Management System.

3.2 Functional Capabilities

CCTV System

- Support for fixed, PTZ, panoramic, and thermal cameras with different resolution.
- Advanced video analytics including motion detection, intrusion detection, people/vehicle counting, facial recognition, license plate recognition (ANPR/LPR), and object left/removed detection.
- Storage management with redundancy (RAID, failover servers).
- Integration with other systems (ACS, Fire Alarm, etc.) for event-based video pop-ups.
- Compliance with ONVIF and RTSP standards.
- Capability to define privacy zones and video masking per regulation.

Access Control System (ACS)

- Multi-credential support (RFID, smart card, mobile NFC, biometric).
- Zone and schedule-based access rules.
- Real-time monitoring and event logging for all entry/exit points.
- Support for anti-passback, two-man rule, and emergency egress.
- Compatibility with fire alarm triggers for automatic door release during emergencies.

Security Management System (SMS)

- Centralized monitoring and management of all connected subsystems (CCTV, ACS, FAS, intrusion alarms).
- Unified alarm and event management dashboard.
- Automatic workflows for incident response and escalation.
- GIS-based map visualization of alarms and live camera feeds.
- Historical incident reporting and root-cause analysis.
- Support for integration with Command & Control Center (C4) for centralized operations.

3.3 Integration and Interoperability

The Integrated Safety and Security Platform shall interface with:

- Building Management System (BMS): For coordinated building-level responses (e.g., lighting or HVAC control during evacuation).
- Fire Alarm System (FAS): For automatic video recording or alarm acknowledgment upon fire events.
- IoT Platform: For event forwarding and data analytics.
- Security Command & Control Center (C4): For real-time visualization, incident management, and analytics.

- Public Address & Voice Alarm System (PAVA): For automatic voice announcements triggered by security events.
- ERP and CAFM Systems: For work order generation and maintenance scheduling.

Integration should use open protocols (ONVIF, BACnet, SNMP, REST API) and data exchange in structured formats (JSON, XML).

All systems must support event correlation and synchronization between alarms, access events, and video evidence.

3.4 Cybersecurity and Data Privacy

- All systems must comply with ISO 27001 and NIST cybersecurity standards.
- Encrypted communication between devices and servers using TLS 1.2/1.3.
- Implementation of role-based access control, multi-factor authentication, and secure audit logging.
- All video and access data to be stored securely with restricted user access.
- Regular vulnerability testing, firmware updates, and penetration testing procedures must be outlined.
- Compliance with local and international privacy laws governing surveillance and biometric data storage.

3.5 User Interface and Experience

- Unified user interface for all systems (CCTV, ACS, SMS) accessible via web client and operator workstation.
- Customizable dashboards for operators, supervisors, and administrators.
- Alarm color coding, filters, and acknowledgment workflows.
- Multi-screen video wall support for Security Operations Center (SOC) display.
- Multi-language support (Arabic/English).

4. Operational and Maintenance Requirements

- Define preventive and corrective maintenance procedures for all system components.
- Provide SLAs for system uptime (minimum 99.9%), incident response, and resolution times.
- Describe monitoring tools used for system health and performance tracking.
- Define periodic testing and calibration routines (camera cleaning, reader calibration, server health checks).
- Provide firmware upgrade policy and lifecycle management procedures.
- Include operator and maintenance staff training programs.
- Provide documentation for incident reporting, root-cause analysis, and escalation hierarchy.

5. Deliverables

MSI shall provide the following deliverables as part of this RFI response:

- Comprehensive system architecture diagram and integration framework.
- Product technical datasheets for CCTV cameras, ACS controllers, and SMS platform.
- Network topology and connectivity plan.
- List of supported analytics and integrations.
- Compliance and certification documentation (ONVIF, UL, ISO, PCI DSS if applicable).
- Reference project details for similar integrated security deployments.
- Operations and maintenance plan including SLAs.

3.4. Use Case 4 – Intelligent Lamp Posts and Smart Poles

1. Background and Objective

As part of the Smart Community vision, the Intelligent Lamp Posts and Smart Poles initiative aims to transform traditional street lighting infrastructure into a multifunctional, connected, and energy-efficient smart urban asset.

The Smart Poles will not only provide adaptive lighting based on environmental conditions but will also serve as a modular platform for hosting multiple smart city applications such as public Wi-Fi, CCTV, environmental sensors, digital signage, and communication antennas (5G, LoRa, etc.).

The objective of this RFI is to collect detailed information from the MSI regarding the proposed architecture, system design, component specifications, and integration approach of the Intelligent Lamp Post and Smart Pole solution, ensuring alignment with the overall Smart Community IoT ecosystem and sustainability targets.

2. Scope of Work

MSI shall design, supply, install, configure, and commission Intelligent Lamp Posts and Smart Poles across the Smart Community, supporting the following functions and features:

- Adaptive LED Street lighting based on motion detection, ambient light levels, and scheduling.
- Remote monitoring and control of luminaires via centralized lighting management software.
- Integrated mounting provisions for:
 - CCTV surveillance cameras.
 - Environmental sensors (air quality, temperature, humidity, noise, etc.).
 - Public Wi-Fi access points and small cell/5G antennas.
 - Digital signage or information displays.
 - EV charging points (where applicable).
- Integration with Smart Community platforms such as IoT, BMS, and Energy Management Systems.
- Compliance with IEC, IEEE, and lighting efficiency standards (e.g., IEC 60598, EN 13201, ISO 50001).
- Design for aesthetic consistency, structural stability, and ease of maintenance.

MSI shall ensure that all components are modular and scalable to accommodate future technologies or service expansions.

3. Technical Requirements

3.1 System Architecture

Provide a detailed architecture diagram showing connections between lamp posts, controllers, communication gateways, and the central management system.

The lighting control system shall support both individual luminaire control and group-based control.

System architecture shall include:

- Smart Controller Unit: Installed within each pole for local intelligence.
- Communication Gateway: To aggregate data from multiple poles and connect to the cloud or central server.
- Central Management Software (CMS): For system-wide control, monitoring, reporting, and analytics.
- Specify communication protocols (e.g., LoRaWAN, Zigbee, NB-IoT, LTE, or wired options).
- Include redundancy and failover provisions for power and communication.
- Define expected system latency, data refresh rates, and power consumption profiles.

3.2 Functional Capabilities

Lighting Control:

- Automatic dimming and brightness adjustment based on motion sensors or ambient light.
- Programmable on/off schedules and event-triggered control.
- Energy consumption tracking and fault detection (lamp failure, power loss).

Environmental Monitoring:

- Integration of environmental sensors for air quality, temperature, humidity, CO₂, and noise.
- Real-time data collection and visualization via dashboards.

Connectivity & Smart Services:

- Integration of public Wi-Fi hotspots and IoT gateways.
- Hosting of surveillance cameras or emergency call boxes.
- Optional integration of EV charging sockets or USB charging ports.

Maintenance & Diagnostics:

- Predictive maintenance based on usage patterns and device performance.
- Automatic alerts for system faults or component failures.

Analytics & Reporting:

- Real-time dashboards for power consumption and fault status.
- Historical data analysis for optimizing energy savings.

3.3 Integration and Interoperability

The Intelligent Lamp Posts and Smart Poles solution shall integrate with:

- IoT Platform: For centralized monitoring and control, using standard APIs and MQTT/REST protocols.
- Energy Management System (EMS): For energy analytics, demand-side optimization, and load management.
- Building Management System (BMS): For integrated facility-level control and status visualization.

- Public Wi-Fi Network: For connectivity to other smart city applications.
- CCTV and Security Systems: For unified event response and power/data integration.

Integration shall comply with open communication standards (BACnet, SNMP, Modbus, OPC UA, etc.) to ensure interoperability with third-party platforms.

3.4 Cybersecurity and Data Privacy

- All data communications between devices, gateways, and central servers must be encrypted using TLS 1.2/1.3 or equivalent.
- Implement role-based authentication and secure access for all management interfaces.
- The system shall comply with ISO 27001 and IEC 62443 for industrial cybersecurity.
- Provide mechanisms for device identity management, firmware integrity verification, and intrusion detection.
- Ensure that no personally identifiable data (e.g., from cameras or Wi-Fi access) is stored without proper anonymization.
- Include procedures for regular penetration testing and vulnerability remediation.

3.5 User Interface and Experience

- Centralized Lighting Management System (LMS) dashboard accessible via web browser and mobile device.
- Real-time status visualization of each pole, lamp, and connected sensor.
- GIS-based map view for device locations, alerts, and performance indicators.
- Customizable dashboards for operations, energy reports, and maintenance analytics.
- Multi-language support (Arabic/English).
- User roles with distinct access privileges (operator, supervisor, administrator).

4. Operational and Maintenance Requirements

- Provide preventive and corrective maintenance procedures for luminaires, sensors, controllers, and communication devices.
- Define SLAs for system uptime ($\geq 99.9\%$) and fault response time.
- Specify lamp lifetime expectancy ($\geq 50,000$ hours) and maintenance intervals.
- Include remote monitoring capabilities for fault detection and predictive maintenance.
- Describe the spare parts and replacement component management strategy.
- Define operator and maintenance team training programs.
- Include sustainability practices such as end-of-life recycling and eco-friendly materials.

5. Deliverables

MSI shall submit the following as part of the RFI response:

- System architecture and communication topology diagrams.
- Product datasheets for poles, luminaires, sensors, and controllers.

- Integration architecture showing data exchange with IoT and EMS systems.
- Power and network design considerations.
- Compliance certificates (IEC, ISO, UL, RoHS).
- Maintenance and operation plan with SLAs.
- Reference projects and manufacturer credentials.

3.5. Use Case 5 – Mobile App (Citizen Engagement Experience App)

1. Background and Objective

As part of the Smart Community digital experience strategy, the Mobile App (Citizen Engagement Experience App) aims to provide an integrated digital gateway that enhances the engagement, convenience, and satisfaction of residents and visitors.

This mobile application will serve as the central interface between residents and the Smart Community's services; enabling seamless access to digital payments, navigation, events, parking and customer support.

The purpose of this RFI is to obtain detailed technical, functional, and operational information from the MSI on the proposed Mobile App solution, ensuring alignment with the Smart Community's overall architecture, data integration strategy, and user experience vision.

2. Scope of Work

MSI shall design, develop, configure, and deploy a Citizen Engagement Experience Mobile Application that serves as the official digital companion for all residents and visitors. The application shall:

- Support ticket purchase, booking, and digital entry through integration with the Electronic Ticketing System and Access Control.
- Enable cashless digital payments through integration with the Digital Payment Platform.
- Offer real-time navigation and indoor/outdoor wayfinding using GPS and BLE beacons.
- Integrate with Parking Management, CCTV (for safety alerts), and IoT sensors for live service status (e.g., queue times, facility availability).
- Support user profiles, loyalty programs, and push notifications for marketing and event updates.
- Allow visitors to submit feedback, requests, or incident reports to the Help Desk system.
- Support both iOS and Android platforms, developed natively or using cross-platform frameworks.
- Ensure compliance with data privacy and accessibility standards.

3. Technical Requirements

3.1 System Architecture

Provide a detailed architecture diagram illustrating integration between mobile app, backend services, APIs, and external systems.

Specify whether the backend will be hosted on-premises, in the cloud, or hybrid, and detail server specifications, databases, and middleware used.

The architecture should include:

- Frontend Mobile App: Native iOS and Android applications.
- Backend Platform: RESTful APIs, databases, authentication, and integration middleware.
- Admin Portal: Web-based management console for content, analytics, and configurations.
- Ensure modularity and scalability to accommodate future features and integrations.

- Outline the proposed security and performance measures (caching, load balancing, CDN).

3.2 Functional Capabilities

The Mobile App shall provide, at minimum, the following modules and features:

User Management and Authentication

- User registration and login via email, phone number, or social media.
- Role-based access control (visitor, resident, operator, admin).
- Integration with Single Sign-On (SSO) and federated identity management.

Ticketing and Booking

- Real-time ticket purchasing and booking integration with the Electronic Ticketing System (Use Case 23).
- QR code-based entry and digital passes stored within the app.
- Secure transaction handling via the Digital Payment Platform (Use Case 1).

Navigation and Wayfinding

- Map-based navigation for the park area using GPS and indoor positioning (BLE, Wi-Fi triangulation).
- Search and filter functionalities (attractions, restrooms, dining, parking).
- Accessibility options for differently-abled users.

Notifications and Marketing

- Personalized push notifications and alerts for events, promotions, and safety announcements.
- Geo-fencing capabilities to trigger notifications based on location.
- Integration with marketing platforms for campaign management.

Customer Support and Feedback

- In-app Help Desk integration (linked to Use Case 20 – Public Help Desk & Call Center).
- Feedback forms, chat support, and issue tracking.
- Integration with CAFM for maintenance-related service tickets.

Analytics and Reporting

- User behavior analytics and heatmaps (visitor flow, dwell time, activity tracking).
- Dashboard for visitor insights and app performance metrics.
- Integration with ERP and CRM systems for business intelligence reporting.

3.3 Integration and Interoperability

The Mobile App shall integrate with the following systems:

- Digital Payment Platform (Use Case 1) – For ticketing and in-app purchases.

- Electronic Ticketing System (Use Case 23) – For event and access management.
- IoT Platform (Use Case 9) – For live operational updates (parking, crowd density, environmental status).
- ERP (Use Case 15) – For user data synchronization and financial records.
- Security Platform (Use Case 3) – For alerts and emergency communication.
- Public Help Desk (Use Case 20) – For issue reporting and service management.

All integrations must use secure APIs (REST/JSON, OAuth 2.0) and standardized data exchange formats.

MSI shall define data flow diagrams, latency expectations, and failover mechanisms.

3.4 Cybersecurity and Data Privacy

- All user data must be handled according to international data protection standards (e.g., GDPR).
- Encrypt all data in transit (TLS 1.2/1.3) and at rest (AES-256).
- Apply strict authentication and authorization policies (multi-factor authentication, session management).
- Implement mobile app security best practices:
 - Code obfuscation and tamper detection.
 - Secure API gateway with rate limiting.
 - Periodic vulnerability and penetration testing.
 - Provide clear user consent and data retention policies.
 - Ensure compliance with ISO 27001 and OWASP Mobile Security standards.

3.5 User Interface and Experience

- The app shall provide a modern, intuitive, and responsive user interface, adhering to UX best practices.
- Follow Apple Human Interface Guidelines and Google Material Design standards.
- Support Arabic and English languages, with right-to-left (RTL) UI support.
- Offer customizable home screens, themes, and shortcuts.
- Include accessibility features (text-to-speech, color contrast, screen reader support).

4. Operational and Maintenance Requirements

- Provide full operational and maintenance support including updates, monitoring, and bug fixes.
- Define Service Level Agreements (SLAs) for app uptime, response time, and support tickets.
- Include processes for version control, feature enhancement, and patch management.
- Provide analytics tools for app performance and crash reporting.
- Offer training for content administrators and technical staff.
- Submit maintenance and release schedules for major and minor updates.
- Ensure 24/7 monitoring of backend services and mobile APIs.

5. Deliverables

MSI shall provide the following as part of the RFI response:

- Detailed system architecture and integration diagram.
- Application wireframes or UI mockups.
- Functional and technical specifications document.
- Integration plan with all relevant Smart Community systems.
- Security and data protection compliance certificates.
- Operation, maintenance, and support plan.
- Reference projects of similar smart community mobile applications.

3.6. Use Case 6 – Smart Waste Management

1. Background and Objective

The Smart Waste Management initiative is a key component of the Smart Community's sustainability and operational efficiency goals. It focuses on leveraging IoT technologies, data analytics, and automation to optimize the collection, transportation, and processing of waste within the community.

This system aims to replace traditional, schedule-based waste collection with an intelligent, data-driven model that monitors waste levels, optimizes collection routes, reduces operational costs, and enhances environmental performance.

The purpose of this RFI is to collect detailed technical, operational, and integration information from the MSI regarding the proposed Smart Waste Management solution, ensuring alignment with the Smart Community's IoT Platform, CAFM, and Environmental Management objectives.

2. Scope of Work

MSI shall design, supply, install, configure, and commission a Smart Waste Management System that includes the following components and functionalities:

- Smart waste bins and compactors equipped with fill-level sensors, temperature sensors, and GPS tracking.
- Wireless communication between bins and the central platform for real-time monitoring.
- Route optimization algorithms for collection vehicles based on bin status and location.
- Integration with Fleet Management Systems for vehicle tracking and scheduling.
- Centralized Waste Management Dashboard for monitoring, analytics, and control.
- Integration with CAFM for work order creation and maintenance requests.
- Support for multi-type waste segregation (recyclable, organic, hazardous, general).
- Compliance with local environmental and waste management regulations.

The solution should ensure operational transparency, improve resource allocation, and minimize the environmental footprint associated with waste collection activities.

3. Technical Requirements

3.1 System Architecture

- Provide a complete system architecture diagram illustrating smart bin sensors, communication networks, gateways, cloud servers, and user dashboards.
- Indicate whether the system will be cloud-hosted, on-premises, or hybrid, and describe data storage and backup strategy.
- The architecture should include:
 - Smart Waste Bins: Equipped with sensors and communication modules.
 - Communication Layer: Using LPWAN (LoRaWAN, NB-IoT, Sigfox) or cellular connectivity.
 - IoT Platform Integration: For data ingestion, analytics, and visualization.

- Central Monitoring Software: For operations and reporting.
- The architecture must support scalability for future expansion (more bins, new collection zones).
- Provide details on device management, firmware updates, and alert mechanisms.

3.2 Functional Capabilities

Smart Bin Monitoring:

- Real-time measurement of waste fill-level, temperature, and bin status.
- Alerts for overflow, fire, or damage.
- Predictive analytics for waste generation patterns.

Route Optimization:

- AI-driven collection scheduling based on real-time data.
- Dynamic route recalculation for operational efficiency.
- Integration with GPS-enabled collection vehicles.

Fleet Management Integration:

- Real-time vehicle tracking, status monitoring, and fuel consumption analytics.
- Automatic dispatching based on bin alerts and proximity.

Reporting and Analytics:

- Dashboard displaying bin occupancy, collection trends, and route efficiency.
- KPI tracking (collection frequency, operational costs, emissions reduction).
- Data export and integration with environmental reports.

Maintenance & Alerts:

- Automatic fault reporting to CAFM for bin or sensor maintenance.
- SMS/email/mobile notifications for abnormal events.

3.3 Integration and Interoperability

The Smart Waste Management System shall integrate with:

- IoT Platform (Use Case 9): For centralized data collection and analytics.
- CAFM (Use Case 2): For maintenance requests, scheduling, and work order management.
- ERP (Use Case 15): For cost tracking, asset inventory, and procurement.
- Energy Management System (Use Case 21): For monitoring fuel consumption and energy efficiency.
- Mobile App (Use Case 5): For public notifications (e.g., waste collection schedules).

Integration shall use open APIs and standardized communication protocols such as MQTT, REST/JSON, or HTTPS.

Data exchange should comply with interoperability standards (IoT schema, ISO 37120 for Smart Cities).

3.4 Cybersecurity and Data Privacy

- Secure all communication channels using TLS 1.2/1.3 or equivalent encryption.
- Implement device authentication and secure firmware management.
- Data at rest must be encrypted using AES-256 or equivalent.
- Ensure compliance with ISO 27001 and IEC 62443 cybersecurity standards.
- Include incident detection and response capabilities (IDS/IPS).
- Define data retention, anonymization, and privacy protection policies for IoT data.
- Provide an audit trail for all device interactions and operator activities.

3.5 User Interface and Experience

- Provide a centralized dashboard with GIS-based visualization of all waste bins and vehicles.
- Enable filtering and grouping by waste type, collection zone, and service status.
- Include KPI widgets for waste volume, route efficiency, and environmental metrics.
- Support multiple user roles (administrator, operations manager, driver, maintenance).
- Mobile-friendly interface for operators and collection teams.
- Multi-language support (Arabic/English).

4. Operational and Maintenance Requirements

- Provide preventive and corrective maintenance plans for sensors, bins, and communication gateways.
- Define SLAs for system uptime ($\geq 99.5\%$), data refresh rates, and issue resolution.
- Include processes for calibration, firmware updates, and remote diagnostics.
- Specify the spare parts management and warranty for all components.
- Define operator training and system handover procedures.
- Include environmental impact monitoring (CO₂ emissions saved, waste diversion rate).
- Provide health, safety, and environmental compliance documentation.

5. Deliverables

MSI shall include the following as part of this RFI response:

- Detailed system architecture and integration diagrams.
- Product datasheets for sensors, bins, communication devices, and gateways.
- Integration and interoperability framework.
- Compliance and certification documents (ISO, IEC, EN standards).
- Operations and maintenance plan.
- Reference projects for smart waste or environmental management systems.
- Proposed roadmap for system deployment and scalability.

3.7. Use Case 7 – Electric Vehicles Charging Stations & Points (EVCS)

1. Background and Objective

The Electric Vehicle Charging Stations & Points (EVCS) initiative forms a key pillar of the Smart Community's sustainability and green mobility strategy. It aims to enable efficient, accessible, and reliable EV charging infrastructure that supports both residents and visitors.

The system should provide scalable, interoperable, and energy-efficient EV charging facilities, fully integrated with the community's Energy Management System (EMS), Building Management System (BMS), and Digital Payment Platform.

The objective of this RFI is to obtain detailed technical, operational, and integration information from the MSI regarding the design, implementation, and management of the EVCS solution, ensuring alignment with international standards and the Smart Community's overall mobility vision.

2. Scope of Work

MSI shall design, supply, install, configure, and commission a comprehensive EV Charging Infrastructure that includes:

- AC and DC charging stations for various vehicle types (public, private, utility).
- Smart chargers supporting load management, dynamic pricing, and energy optimization.
- Integration with the Energy Management System (EMS) for grid balancing and power analytics.
- Integration with Digital Payment Platform (Use Case 1) for user-friendly cashless payments.
- Centralized EV Charging Management Software (CMS) for monitoring and control.
- User access via mobile app and RFID/NFC cards.
- Compliance with IEC 61851, ISO 15118, and OCPP (Open Charge Point Protocol).
- Support for future expansion (additional chargers or fast-charging hubs).

The scope includes design, procurement, installation, testing, commissioning, and integration with the Smart Community's core systems.

3. Technical Requirements

3.1 System Architecture

- Provide a detailed architecture showing charger units, communication gateways, CMS servers, and interfaces with EMS and BMS.
- The system must support both local and cloud-based management.
- Include failover, redundancy, and remote diagnostics capabilities.
- All components must comply with relevant standards for electrical safety, cybersecurity, and energy efficiency.
- Specify network communication protocols (OCPP 1.6/2.0.1, Modbus, or equivalent).
- The CMS should allow real-time status visualization, power flow tracking, and session management.

3.2 Functional Capabilities

Charging Operations:

- Support for AC (Type 2) and DC fast charging (CCS2, CHAdeMO).
- Variable charging power levels (7kW to 350kW).
- Automatic session start/stop via RFID, mobile app, or QR code.
- Load balancing across multiple stations to prevent grid overload.

Energy Optimization:

- Integration with EMS for demand response and peak shaving.
- Smart charging schedules based on energy tariffs and grid capacity.
- Integration with renewable energy sources (e.g., solar PV).

User Management:

- User authentication via RFID/NFC or mobile credentials.
- Support for registered users, guests, and fleet accounts.
- Dynamic pricing models (time-based, energy-based, or flat rate).

Monitoring & Analytics:

- Real-time monitoring of charging sessions, voltage, and power consumption.
- Alerts for faults, overcurrent, or communication errors.
- Historical reports and predictive analytics for charger usage.

Maintenance:

- Automatic fault reporting to CAFM (Use Case 2).
- Remote firmware upgrades and diagnostics.

3.3 Integration and Interoperability

The EVCS platform shall integrate with the following systems:

- Energy Management System (Use Case 21): For load management and energy optimization.
- Building Management System (Use Case 19): For monitoring power consumption and control.
- Digital Payment Platform (Use Case 1): For billing and transactions.
- IoT Platform (Use Case 9): For device data ingestion and analytics.
- Mobile App (Use Case 5): For user registration, navigation to nearest station, and session control.
- ERP System (Use Case 15): For financial reconciliation and reporting.

Integration should follow open standards (OCPP, MQTT, REST APIs, or Modbus TCP/IP).

The system should ensure data synchronization between subsystems and centralized reporting.

3.4 Cybersecurity and Data Privacy

- Compliance with ISO 27001, IEC 62443, and NIST standards for cybersecurity.
- Encrypted communication between chargers, CMS, and integrated systems (TLS 1.2/1.3).
- Authentication and access control for operators, technicians, and users.
- Protection of user payment data in line with PCI DSS requirements.
- Regular vulnerability scanning, patch management, and firmware security validation.
- Secure storage and anonymization of user and vehicle data.

3.5 User Interface and Experience

- Centralized dashboard for operators to monitor all stations and active sessions.
- GIS map view showing station locations, occupancy, and charger availability.
- Mobile app integration for users (start/stop sessions, view history, make payments).
- Multi-language support (Arabic/English).
- Intuitive UI for both administrative and customer-facing interfaces.

4. Operational and Maintenance Requirements

- Provide preventive and corrective maintenance schedules for all charging stations and CMS components.
- Define SLAs for charger uptime ($\geq 99.5\%$), response times, and fault resolution.
- Include remote diagnostics, self-healing, and automated alerts.
- Provide training for operators, technicians, and administrators.
- Include procedures for firmware and hardware updates.
- Provide warranty and post-commissioning support documentation.
- Implement regular system performance reporting and KPI tracking (usage rate, energy output, CO₂ savings).

5. Deliverables

MSI shall provide the following as part of this RFI response:

- Detailed system and network architecture diagrams.
- Technical datasheets for EV chargers, gateways, and CMS platform.
- Integration plan and data flow diagram.
- Compliance certificates (OCPP, IEC, ISO, UL).
- Operations and maintenance plan including SLAs.
- Cybersecurity framework and data protection policies.
- Reference projects with similar EV infrastructure.

3.8. Use Case 8 – Security Command & Control Center / Operations Control Room

1. Background and Objective

The Security Command & Control Center (C4 Center) represents the central nervous system of the Smart Community's safety, operations, and situational awareness ecosystem. It provides a unified command environment that consolidates data from multiple subsystems including CCTV, Access Control, Fire Alarm, IoT sensors, and Building Systems into one central monitoring and response platform.

The purpose of this RFI is to collect detailed information from the MSI about the proposed design, technologies, and integration framework of the C4 Center, ensuring alignment with the Smart Community's operational strategy and digital transformation objectives.

The C4 Center shall be designed as a 24/7 mission-critical facility, providing operators, supervisors, and management teams with real-time visualization, analytics, and control capabilities to enhance situational awareness and emergency response coordination.

2. Scope of Work

MSI shall design, supply, install, and commission a fully integrated C4 Center that serves as the central hub for the following systems:

- Integrated Safety & Security Platform (Use Case 3)
- Building Management Systems (Use Case 19)
- IoT Platform (Use Case 9)
- Energy Management System (Use Case 21)
- Smart Lighting (use case 4), Waste Management (use case 6), and Irrigation Systems (use case 16)
- Public Address and Voice Alarm System (Use Case 13)
- Public Help Desk & Call Center (Use Case 20)

The scope covers:

- Facility Design: Command room layout, operator consoles, video wall setup, network infrastructure, and ergonomic design.
- Software Integration: Unified monitoring platform consolidating events, alarms, and analytics from all subsystems.
- Incident Management: Real-time event correlation, alarm prioritization, and response workflows.
- Visualization: GIS-based maps, dashboards, 3D models, and real-time analytics.
- Operations Management: Shift scheduling, reporting, and system logging.
- Business Continuity: High-availability architecture, failover redundancy, and disaster recovery.

3. Technical Requirements

3.1 System Architecture

- Provide an architecture diagram showing system layers (data acquisition, middleware, application, visualization).

- The design shall include:
 - Primary Command Center (Main C4).
 - Backup/Disaster Recovery Site (hot standby).
 - Central Data Integration Layer (middleware) to aggregate data from all subsystems.
- System architecture must support real-time, event-driven operations with low latency (<2 seconds).
- Use of virtualized servers or cloud infrastructure for scalability and redundancy.
- Support for modular expansion to include future systems or service areas.
- Ensure network segregation (operational vs. corporate network) with strict security zoning.

3.2 Functional Capabilities

Command & Control Operations

- Real-time monitoring and alarm management for all connected systems.
- Event correlation engine to detect multi-system incidents (e.g., access breach + fire alarm + camera alert).
- Automated incident workflows with predefined SOPs and escalation rules.
- Integration with CCTV for visual verification of alarms.
- Live dashboards and performance KPIs.

Visualization & Reporting

- GIS-based graphical representation of all assets, buildings, and zones.
- 3D digital twin visualization for facilities.
- Custom dashboards per department (security, maintenance, energy).
- Historical event playback and reporting tools.
- Scheduled and on-demand reporting (PDF, Excel).

Communications & Coordination

- Integration with Public Address and Voice Alarm Systems (PAVA) for emergency announcements.
- Integration with Radio Communication or Unified Communication Systems for field coordination.
- Audio-visual call logging and recording.
- Interfacing with emergency services and management authorities.

Analytics & Decision Support

- Data analytics tools for trend identification and predictive maintenance.
- Integration with AI-based anomaly detection and incident forecasting modules.
- Machine learning algorithms for operational insights and optimization.
- Dashboard for KPI measurement (response time, incident closure rate, system uptime).

3.3 Integration and Interoperability

The C4 Center shall act as the central integration layer for all Smart Community systems, including:

- Integrated Security Platform (Use Case 3)
- Fire Alarm System (Use Case 18)
- BMS (Use Case 19)
- EMS (Use Case 21)
- IoT Platform (Use Case 9)
- Public Help Desk (Use Case 20)
- Smart Lighting (Use Case 4)
- Smart Waste Management (Use Case 6)

Integration shall be achieved through open standards and APIs (REST, MQTT, OPC UA, BACnet, ONVIF).

The middleware must support data normalization, alarm correlation, and unified dashboards across all connected systems.

3.4 Cybersecurity and Data Privacy

- Comply with ISO 27001, NIST, and IEC 62443 frameworks.
- Implement multi-layered security architecture (network, application, and endpoint).
- Data encryption (AES-256), secure communication (TLS 1.3).
- Role-based access control and multi-factor authentication.
- Secure logging, audit trails, and tamper-proof incident records.
- Regular penetration testing, vulnerability management, and patching.
- Implementation of Security Information and Event Management (SIEM) integration.

3.5 User Interface and Experience

- Centralized and unified interface for all subsystems.
- Configurable dashboards per operator and function.
- GIS-based visualization, alarm color coding, and live video embedding.
- Multi-monitor and video wall support.
- Arabic/English language interface.
- User profiles (operator, supervisor, manager) with permission hierarchies.

4. Operational and Maintenance Requirements

- Provide maintenance plans for hardware (video walls, consoles) and software (middleware, applications).
- Define SLAs for system uptime ($\geq 99.9\%$), response time, and recovery.
- Establish system health monitoring and alert mechanisms.
- Procedures for periodic software updates, backups, and failover testing.
- Provide operational training for operators, administrators, and maintenance teams.
- Document all SOPs, incident workflows, and escalation processes.
- Include lifecycle management and warranty support for at least 5 years.

5. Deliverables

MSI shall include the following as part of this RFI response:

- Complete C4 Center architecture and design layout (floor plan, network, power, system).
- List of integrated systems and proposed middleware solution.
- Product datasheets for hardware (video wall, workstations, consoles).
- Integration and interoperability documentation.
- Cybersecurity and data protection framework.
- O&M plan, training program, and system documentation.
- Reference projects for command and control centers of similar scale.

3.9. Use Case 9 – IoT Platform (Central IoT Management)

1. Background and Objective

The IoT Platform (Central IoT Management) is the backbone of the Smart Community's digital ecosystem, designed to integrate, manage, and analyze data from thousands of connected devices, sensors, and subsystems deployed across the entire development.

This platform will enable real-time data collection, device management, event processing, and analytics, forming the core of intelligent operations within the Smart Community. It will serve as the foundation for interoperability between all smart systems, including lighting, energy, waste management, security, and environmental monitoring.

The objective of this RFI is to gather comprehensive information from the MSI regarding the architecture, capabilities, standards, and integration approach of the proposed IoT Platform, ensuring it supports scalability, cybersecurity, and interoperability across all Smart Community applications.

2. Scope of Work

MSI shall design, supply, implement, configure, and commission an IoT Platform capable of:

- Centralized management of all IoT devices and sensors across the Smart Community.
- Real-time data acquisition, normalization, and visualization from heterogeneous sources.
- Integration with vertical systems such as Smart Lighting, Smart Waste, BMS, EMS, and CAFM.
- Support for edge computing, data aggregation, and analytics.
- Device provisioning, configuration, and lifecycle management.
- Centralized alarm and event management.
- Data storage, processing, and dashboard generation.
- Support for both cloud and on-premises deployment models.

The platform shall ensure interoperability, data integrity, and security, while enabling advanced analytics and decision-making.

3. Technical Requirements

3.1 System Architecture

- Provide a detailed architecture diagram including devices, edge gateways, IoT brokers, data ingestion layers, and applications.
- Define communication models (device-to-cloud, edge-to-cloud, and cloud-to-cloud).
- Support both real-time streaming and batch data processing.
- Include data lake or centralized database for storing structured and unstructured data.
- Ensure horizontal scalability to accommodate future expansion.
- Include redundancy, load balancing, and disaster recovery mechanisms.
- Support integration with AI/ML engines for predictive analytics.

3.2 Functional Capabilities

Device Management:

- Onboarding, provisioning, and monitoring of IoT devices.
- Remote firmware updates and diagnostics.
- Health monitoring, fault detection, and reporting.

Data Management:

- Data normalization, filtering, and cleansing.
- Real-time event stream processing.
- Long-term data retention policies.
- Support for MQTT, CoAP, HTTP, AMQP protocols.

Analytics and Visualization:

- Real-time dashboards and custom KPI creation.
- Trend analysis and anomaly detection.
- Integration with third-party BI tools (Power BI, Tableau, etc.).
- Rules Engine:
 - Configurable automation rules for event-based actions (e.g., trigger alarm, adjust lighting).
 - Workflow creation for inter-system responses.

APIs and Integration:

- RESTful APIs for data access and system integration.
- SDKs for third-party developers.
- Data export in standard formats (JSON, XML, CSV).

3.3 Integration and Interoperability

The IoT Platform shall act as the central integration hub for the Smart Community, interfacing with:

- Smart Lighting (Use Case 4)
- Smart Waste Management (Use Case 6)
- EV Charging System (Use Case 7)
- BMS (Use Case 19)
- EMS (Use Case 21)
- Security and C4 Systems (Use Case 8)
- Mobile App (Use Case 5)
- CAFM (Use Case 2)
- ERP and Data Analytics Platforms (Use Case 15)

Integration shall use open communication protocols (MQTT, REST, OPC UA, BACnet, SNMP, LoRaWAN).

Data exchange must comply with IoT interoperability standards such as oneM2M, ISO/IEC 30141, and IEEE P2413.

The system should support northbound and southbound integrations to enable bidirectional communication.

3.4 Cybersecurity and Data Privacy

- Comply with ISO 27001, IEC 62443, and NIST Cybersecurity Framework.
- Implement end-to-end encryption for all data in transit and at rest (TLS 1.3, AES-256).
- Support device-level identity and mutual authentication.
- Use of secure key management and digital certificates (PKI).
- Role-based access control, MFA, and audit logging.
- Integration with Security Information and Event Management (SIEM) for threat detection.
- Data anonymization and retention policies in line with GDPR or local privacy laws.

3.5 User Interface and Experience

- Unified web-based dashboard for system administrators, operators, and analysts.
- Real-time visualization of device health, telemetry, and alerts.
- GIS map integration for spatial data visualization.
- Customizable widgets and KPI dashboards.
- Multi-language support (Arabic/English).
- Mobile-responsive UI for remote monitoring.

4. Operational and Maintenance Requirements

- Define maintenance schedules for software components, gateways, and databases.
- Provide SLAs for uptime ($\geq 99.9\%$), latency, and issue resolution.
- Include system monitoring tools and alert mechanisms.
- Define procedures for firmware updates, patches, and performance tuning.
- Provide user training for system administrators and operators.
- Include warranty and long-term technical support documentation.
- Submit O&M plan including periodic audits and performance reviews.

5. Deliverables

MSI shall include the following as part of the RFI response:

- Full IoT Platform architecture diagrams (logical and physical).
- Product datasheets and specifications for IoT gateways and software components.
- API documentation and integration approach.
- Cybersecurity and compliance certifications.
- Operations and maintenance plan.
- Reference projects demonstrating IoT platform deployment at scale.

3.10. Use Case 10 – Integrated Corporate / Operations Billing System

1. Background and Objective

The Integrated Corporate / Park Operations Billing System is a critical component of the Smart Community's financial and operational management ecosystem. It ensures accurate, automated, and transparent billing for all services, utilities, and operations across the park, including tenant services, facility usage, energy consumption, and digital transactions.

This system will enable seamless integration with other Smart Community platforms such as the ERP, IoT, EMS, and Digital Payment Platform to support end-to-end financial workflows from data collection to invoicing and reporting.

The objective of this RFI is to gather detailed information from the MSI regarding the proposed billing system architecture, functionality, and integration framework to ensure alignment with the community's digital and operational strategy.

2. Scope of Work

MSI shall design, supply, install, and commission an Integrated Billing System capable of managing financial and usage data across multiple service domains. The system shall:

- Handle multi-service billing (utilities, parking, facility rentals, digital services, maintenance, etc.).
- Support multi-entity and multi-currency environments.
- Integrate with ERP (Use Case 15), IoT Platform (Use Case 9), and Digital Payment Platform (Use Case 1).
- Provide a centralized dashboard for billing, reconciliation, and financial reporting.
- Support both prepaid and postpaid customer models.
- Enable real-time data collection from sensors, and subsystems.
- Comply with local taxation and regulatory requirements (e.g., VAT).
- Support flexible pricing, discounting, and revenue-sharing models.

The system shall ensure data accuracy, transparency, and auditability across all transactions.

3. Technical Requirements

3.1 System Architecture

- Provide a detailed architecture showing integration between billing engine, data sources, and ERP systems.
- Define whether the system will be cloud-based, on-premises, or hybrid.
- Include layers for data ingestion, billing logic, financial reconciliation, and reporting.
- The architecture must support:
 - Scalability to accommodate large volumes of transactions.
 - Redundancy and failover for critical operations.
 - Secure API connectivity to other smart systems.

- Use of relational or NoSQL databases for high-speed processing.
- Clearly define backup and disaster recovery mechanisms.

3.2 Functional Capabilities

Billing and Rating Engine:

- Automated billing based on consumption or usage data.
- Real-time rating for various service types (utilities, access, parking, etc.).
- Support for time-of-use pricing and dynamic tariffs.

Data Collection and Integration:

- Integration with IoT sensors and smart meters for usage data.
- Automated data import from EMS, BMS, and CAFM.
- Validation and error handling for inconsistent data.

Invoice Management:

- Customizable invoice templates.
- Multi-currency and tax-compliant invoices.
- Email/SMS notifications to customers.

Payment and Reconciliation:

- Integration with Digital Payment Platform for online transactions.
- Automated reconciliation between payments, invoices, and accounting records.
- Refund and adjustment handling.

Analytics and Reporting:

- KPI dashboards (revenue, outstanding payments, usage trends).
- Historical reporting and audit logs.
- Integration with BI tools for predictive financial analytics.

3.3 Integration and Interoperability

The Billing System shall integrate seamlessly with:

- ERP System (Use Case 15) – for general ledger, accounts receivable/payable, and financial reporting.
- Digital Payment Platform (Use Case 1) – for online transactions and reconciliation.
- IoT Platform (Use Case 9) – for consumption and service usage data.
- Energy Management System (Use Case 21) – for meter readings and energy cost allocation.
- CAFM (Use Case 2) – for maintenance and service charge billing.

Integration shall follow open API standards (REST, SOAP, JSON) with secure communication (TLS 1.2/1.3).

Support for ETL (Extract, Transform, Load) processes to ensure accurate data synchronization across systems.

3.4 Cybersecurity and Data Privacy

- Compliance with ISO 27001, PCI DSS, and relevant financial data protection laws.
- Encryption of financial data in transit and at rest.
- Role-based access control with audit trails for all billing transactions.
- Secure API gateways and key management systems.
- Regular vulnerability scanning and patching.
- Ensure segregation between operational and financial data environments.
- Implement data backup and archival procedures in compliance with local regulations.

3.5 User Interface and Experience

- Web-based dashboard for finance and operations teams.
- Configurable views for billing status, revenue tracking, and customer profiles.
- Multi-language interface (Arabic/English).
- Interactive charts and KPI widgets for analytics.
- Responsive design for desktop, tablet, and mobile use.
- User access hierarchy (operator, manager, administrator).

4. Operational and Maintenance Requirements

- Provide procedures for software updates, patching, and data backup.
- Define SLAs for system uptime ($\geq 99.9\%$), processing speed, and support.
- Include database maintenance and log management routines.
- Provide user and administrator training programs.
- Include documentation for integration, configuration, and troubleshooting.
- Provide warranty, support, and version control plan.
- Ensure data reconciliation validation and audit readiness.

5. Deliverables

MSI shall include the following as part of the RFI response:

- System architecture and data flow diagrams.
- Product datasheets and software specifications.
- API documentation for integration with external systems.
- Compliance and certification documents (ISO, PCI DSS).
- O&M and support plan.
- Example reports and dashboards.
- Reference implementations in similar smart community or corporate billing environments.

3.11. Use Case 11 – Building Information Modelling (BIM)

1. Background and Objective

The Building Information Modelling (BIM) initiative is a foundational component of the Smart Community's digital infrastructure strategy. It aims to establish a comprehensive digital representation of physical and functional characteristics of all built assets supporting design coordination, construction management, asset operation, and lifecycle optimization.

The BIM environment will serve as a centralized source of truth for all stakeholders (designers, contractors, operators, and facility managers), enabling efficient collaboration, clash detection, maintenance planning, and data integration with downstream systems such as CAFM, BMS, and IoT Platforms.

The objective of this RFI is to collect detailed information from the MSI regarding the proposed BIM framework, software tools, standards, integration capabilities, and data management approach to ensure alignment with the Smart Community's digital twin and asset management vision.

2. Scope of Work

MSI shall design, implement, and manage a BIM-enabled environment that covers the entire asset lifecycle from design and construction through operation and maintenance. The scope shall include:

- Creation and maintenance of federated BIM models for all facilities and infrastructure.
- Development of BIM Execution Plan (BEP) covering modelling standards, data exchange, and collaboration protocols.
- Integration of BIM data with CAFM, BMS, EMS, and IoT systems.
- Support for Digital Twin implementation and real-time synchronization with IoT data.
- Establishment of Common Data Environment (CDE) for document and model management.
- Implementation of asset tagging and classification (aligned with ISO 19650 and COBie standards).
- Delivery of as-built BIM models for all completed structures.
- Provision of training and documentation for operators and facility management teams.

3. Technical Requirements

3.1 System Architecture

Provide an overview of the BIM system architecture, including modelling tools, databases, and integration layers.

The BIM environment should include:

- Authoring Tools: Autodesk Revit, Bentley AECOsim, or equivalent.
- Collaboration Platform: Common Data Environment (CDE) hosted on-premises or in the cloud.
- Integration Layer: APIs and middleware for linking BIM data with other systems.
- Ensure version control, access management, and audit logging for all data exchanges.
- Provide infrastructure requirements for data hosting, backup, and redundancy.

- Support for open standards such as IFC (Industry Foundation Classes) and COBie for interoperability.

3.2 Functional Capabilities

Design and Coordination:

- 3D model creation for architectural, structural, and MEP disciplines.
- Clash detection and design conflict resolution.
- Construction sequencing (4D) and cost estimation (5D).

Asset Data Management:

- Asset tagging and classification with attributes (serial number, location, maintenance schedule).
- Integration with CAFM and BMS for operational data linkage.
- As-built updates and change tracking.

Collaboration and Review:

- Multi-user access with role-based permissions.
- Review workflows for design validation and approvals.
- Commenting, markup, and version comparison.

Visualization and Analysis:

- Real-time visualization of models in 3D, 4D, and 5D environments.
- Support for immersive visualization (AR/VR integration).
- Reporting and dashboard generation.

Digital Twin Integration:

- Real-time synchronization with IoT sensors via the IoT Platform (Use Case 9).
- Live monitoring of environmental and operational parameters.
- Predictive analytics for asset performance optimization.

3.3 Integration and Interoperability

The BIM system shall integrate with:

- CAFM (Use Case 2): For maintenance scheduling and asset lifecycle management.
- BMS (Use Case 19): For linking building elements to real-time operational data.
- EMS (Use Case 21): For mapping energy performance metrics to building systems.
- IoT Platform (Use Case 9): For two-way synchronization of live sensor data.
- EDMS (Use Case 12): For document and drawing management.
- ERP (Use Case 15): For procurement and cost tracking.

Integration shall be achieved through open APIs, IFC, and COBie data formats, ensuring full interoperability across platforms.

The solution should comply with ISO 19650, PAS 1192, and relevant BIM maturity standards.

3.4 Cybersecurity and Data Privacy

- All BIM data exchanges and storage must be encrypted using TLS 1.2/1.3 and AES-256.
- Implement strict user authentication, role-based access control, and audit trails.
- Comply with ISO 27001 and ISO 19650 Part 5 (Security-minded BIM).
- Regular vulnerability and compliance audits of BIM servers and CDE environments.
- Define secure data retention and archival policies.
- Ensure intellectual property protection and data access controls for all stakeholders.

3.5 User Interface and Experience

- Provide intuitive web-based and desktop interfaces for viewing and managing BIM models.
- Support for multi-discipline collaboration in real time.
- Integrate GIS and IoT data visualization layers.
- Support for both Arabic and English languages.
- Enable customizable dashboards and reports for management and facility teams.

4. Operational and Maintenance Requirements

- Provide SLAs for system uptime ($\geq 99.9\%$) and model access performance.
- Include preventive and corrective maintenance schedules for BIM tools and CDE.
- Provide procedures for data backup, restoration, and version control.
- Conduct periodic model validation and quality assurance reviews.
- Offer training and certification programs for BIM coordinators and FM users.
- Include warranty and long-term technical support.
- Submit documentation for BEP, naming conventions, and model handover.

5. Deliverables

MSI shall include the following as part of this RFI response:

- BIM system architecture and integration diagrams.
- BIM Execution Plan (BEP) outline.
- Standards compliance documentation (ISO 19650, COBie, IFC).
- Example BIM models and CDE screenshots.
- Integration framework and API documentation.
- O&M and training plan.
- Reference projects demonstrating BIM and digital twin implementations.

3.12. Use Case 12 – Electronic Document Management System (EDMS)

1. Background and Objective

The Electronic Document Management System (EDMS) is a core enabler of the Smart Community's digital transformation and governance framework. It is intended to provide a centralized, secure, and scalable platform for managing, storing, sharing, and controlling all project and operational documents throughout their lifecycle from creation to archival.

The system will serve as the single source of truth for all stakeholders, ensuring document version control, accessibility, traceability, and regulatory compliance. It will also act as a foundation for digital workflows and collaboration across departments and service providers.

The objective of this RFI is to obtain detailed technical and functional information from the MSI on the proposed EDMS solution, its architecture, integration capabilities, and operational approach to ensure consistency, efficiency, and security in document management across the Smart Community.

2. Scope of Work

MSI shall design, supply, configure, and commission an Electronic Document Management System that provides:

- Centralized repository for all digital documents (design files, contracts, reports, manuals, correspondence).
- Advanced search, categorization, and indexing features.
- Role-based access control and approval workflows.
- Version control and audit trail functionality.
- Support for document lifecycle management (draft → review → approved → archive).
- Integration with email, ERP, BIM, CAFM, and project management platforms.
- Support for digital signatures and metadata tagging.
- Multi-language user interface and localization support.
- Compliance with ISO 9001 and ISO 15489 records management standards.

3. Technical Requirements

3.1 System Architecture

- Provide a detailed architecture including application servers, database servers, storage, and user access layers.
- Define whether the system will be cloud-based, on-premises, or hybrid.
- Include modules for:
 - Document Repository
 - Workflow Engine
 - Search and Metadata Management
 - Security and Compliance
- Ensure scalability to accommodate multiple departments and large file volumes.

- Include disaster recovery, redundancy, and data backup features.
- Specify the database type (SQL/NoSQL) and integration framework.

3.2 Functional Capabilities

Document Management:

- Upload, categorize, and tag documents using metadata fields.
- Automated document numbering and versioning.
- Full-text search, OCR, and indexing capabilities.

Workflow Automation:

- Configurable workflows for document review and approval.
- Integration with email notifications and task assignments.

Access Control:

- Role- and department-based permissions.
- Secure external sharing with time-limited access.

Collaboration and Integration:

- Real-time co-authoring and commenting.
- Integration with BIM (Use Case 11), CAFM (Use Case 2), and ERP (Use Case 15).
- API-based interoperability with other Smart Community platforms.

Archiving and Retention:

- Automated retention policies and archival schedules.
- Long-term storage in compliance with legal and organizational requirements.

Reporting and Analytics:

- Dashboard for document statistics, pending approvals, and user activity.
- KPI monitoring for document turnaround times and workflow performance.

3.3 Integration and Interoperability

The EDMS shall integrate with:

- BIM Platform (Use Case 11): For storing design drawings and revision histories.
- CAFM (Use Case 2): For document-based maintenance procedures.
- ERP (Use Case 15): For invoices, purchase orders, and financial records.
- IoT Platform (Use Case 9): For attaching documentation to connected assets.
- Public Help Desk (Use Case 20): For service request and incident documentation.

Integration shall follow open standards (REST API, SOAP, JSON, XML).

Support for interoperability with Microsoft Office 365, SharePoint, and AutoCAD file formats is required.

3.4 Cybersecurity and Data Privacy

- Compliance with ISO 27001 and NIST cybersecurity standards.
- Data encryption at rest (AES-256) and in transit (TLS 1.2/1.3).
- Implementation of user authentication (Active Directory, SSO, MFA).
- Full audit logging of user actions (upload, download, delete, share).
- Secure data retention and deletion policies aligned with local regulations.
- Regular vulnerability scans, patch management, and security audits.
- Ensure intellectual property protection for project and client documents.

3.5 User Interface and Experience

- Web-based responsive interface accessible from desktop and mobile devices.
- User-friendly dashboard with intuitive navigation and search functions.
- Support for Arabic and English languages.
- Role-based home screens for administrators, document controllers, and users.
- Drag-and-drop upload, batch processing, and document preview.

4. Operational and Maintenance Requirements

- Provide SLAs for uptime ($\geq 99.9\%$) and document retrieval performance.
- Define preventive and corrective maintenance plans.
- Include backup and restoration procedures for document repositories.
- Provide administrator and end-user training.
- Include version upgrades, patch management, and long-term technical support.
- Define archival verification and disaster recovery testing schedules.

5. Deliverables

MSI shall include the following as part of the RFI response:

- System architecture and integration diagrams.
- Product datasheets and technical specifications.
- Workflow and access control examples.
- Compliance certificates (ISO 9001, ISO 15489, ISO 27001).
- Operations and maintenance plan.
- User training and support documentation.
- Reference projects for similar EDMS implementations.



3.13. Use Case 13 – Public Address & Voice Alarm Systems (PAVA)

1. Background and Objective

The Public Address and Voice Alarm (PAVA) System plays a vital role in the Smart Community's safety, emergency communication, and operational notification infrastructure.

It is designed to ensure that clear, reliable, and timely audio messages are broadcast across all public areas, buildings, and facilities for both daily operational announcements and emergency alerts.

The system will form part of the ICT and safety integration ecosystem, interfacing with fire alarm, BMS, C4 Center, and IoT systems to enable automated, event-driven responses.

The objective of this RFI is to obtain detailed technical and design information from the MSI regarding the PAVA solution architecture, integration approach, and operational features to ensure full compliance with safety standards and interoperability across the Smart Community's digital platforms.

2. Scope of Work

MSI shall design, supply, install, and commission a Public Address and Voice Alarm (PAVA) System covering:

- All public areas, buildings, and critical infrastructure across the Smart Community.
- Integration with fire alarm systems for emergency voice evacuation.
- Integration with C4 Command & Control Center for centralized operation and monitoring.
- Integration with BMS and IoT Platform for automatic alerts and announcements.
- Multi-zone audio distribution with redundancy and fault tolerance.
- Centralized and local control capabilities.
- High-quality, intelligible voice reproduction and background music functionality.
- Compliance with EN 54-16, EN 54-24, and other relevant standards.

3. Technical Requirements

3.1 System Architecture

Provide a detailed system block diagram showing central equipment, amplifiers, speaker lines, and control units.

Architecture shall include:

- Central Control Rack at the main data center or C4 Center.
- Local Zone Equipment in each building or cluster.
- Redundant Network Backbone for audio distribution and control.
- Integration Interfaces with Fire Alarm and C4 systems.

The system shall support:

- IP-based audio transmission over redundant Ethernet networks.

- Centralized monitoring and diagnostics.
- Scalability to add future zones or buildings.
- Distributed amplifiers and loudspeakers with line monitoring.

3.2 Functional Capabilities

Public Address Functions:

- Background music and paging capability.
- Multi-zone broadcasting and individual zone control.
- Scheduled and on-demand announcements.
- Integration with the Building Management System for operational messages.

Voice Alarm Functions:

- Automatic evacuation messages triggered by the Fire Alarm System (Use Case 18).
- Pre-recorded and live message broadcasting.
- Priority override for emergency announcements.
- Dual power supply and line monitoring for system reliability.

Control and Monitoring:

- Centralized operation via a web-based or dedicated control console.
- Real-time fault monitoring and event logs.
- Integration with C4 Center for remote supervision and incident coordination.

Audio Quality and Coverage:

- Minimum speech intelligibility rating (STI ≥ 0.5).
- Compliance with acoustic and environmental design standards.
- Support for various speaker types (ceiling, wall-mounted, horn, column).

3.3 Integration and Interoperability

The PAVA system shall integrate seamlessly with:

- Fire Alarm System (Use Case 18) – automatic emergency voice alarms.
- C4 Center (Use Case 8) – central monitoring and manual broadcast control.
- BMS (Use Case 19) – automated alerts based on system alarms or conditions.
- IoT Platform (Use Case 9) – event-triggered notifications and analytics.
- Public Help Desk (Use Case 20) – for incident and information announcements.

Integration shall utilize open communication protocols (TCP/IP, SIP, or OPC).

System APIs must allow bi-directional data exchange with supervisory platforms.

All network interfaces must comply with ICT standards and VLAN segmentation requirements.

3.4 Cybersecurity and Data Privacy

- System design must adhere to IEC 62443 and ISO 27001 standards for secure communication.
- Audio transmission shall be encrypted over IP (TLS 1.3).
- Implement secure user authentication and access control for operators.
- Maintain audit logs for all system actions (paging, message broadcast, faults).
- Isolate the PAVA network within the ICT infrastructure using firewalls and VLANs.
- Perform regular cybersecurity vulnerability assessments and patch management.

3.5 User Interface and Experience

- Intuitive, multilingual control interface (Arabic/English).
- Graphical zoning map with live status indicators.
- User roles (Operator, Supervisor, Administrator) with defined privileges.
- Support for touchscreen operation and mobile access.
- Central dashboard showing alarms, faults, and broadcast history.
- Integration with video walls at the C4 Center for visual status display.

4. Operational and Maintenance Requirements

- Define SLAs for system uptime ($\geq 99.9\%$) and audio availability.
- Include preventive maintenance schedule for amplifiers, speakers, and network components.
- Provide O&M manuals and as-built drawings.
- Include procedures for system testing, speaker line monitoring, and fault diagnosis.
- Provide training for system operators and maintenance staff.
- Ensure spare parts availability and technical support for at least 5 years.
- Submit a warranty and lifecycle support plan.

5. Deliverables

MSI shall include the following as part of the RFI response:

- Complete system architecture and network topology diagrams.
- Product datasheets and certifications (EN 54-16, EN 54-24).
- Integration plan with Fire Alarm and C4 Center.
- Audio coverage design and intelligibility calculations.
- Cybersecurity compliance documents.
- Operations and maintenance plan.
- Reference projects demonstrating large-scale PAVA installations.

3.14. Use Case 14 – Core, Consolidated Smart Cities Platform

1. Background and Objective

The Core, Consolidated Smart Cities Platform (SCP) is the central digital backbone of the entire Smart Community, integrating data, analytics, and control functions across all smart domains including mobility, utilities, energy, safety, environment, and citizen engagement.

This platform is designed to unify all vertical systems under a single, interoperable, and secure framework, providing real-time situational awareness, operational intelligence, and decision support to city operators and management teams.

The objective of this RFI is to gather comprehensive details from the MSI regarding the proposed Smart Cities Platform architecture, interoperability mechanisms, cybersecurity posture, and scalability approach.

The solution must ensure seamless data flow between all Smart Community systems, enabling centralized command, monitoring, and optimization of services.

2. Scope of Work

MSI shall design, implement, and configure a Core Smart Cities Platform that consolidates and orchestrates all smart systems under one unified environment. The platform shall:

- Integrate all use cases defined within the Smart Community scope (IoT, BMS, EMS, PAVA, CCTV, Access Control, Waste, Irrigation, etc.).
- Provide a central data integration layer for aggregation, normalization, and analytics.
- Offer real-time dashboards, KPI tracking, and AI-based insights for operational efficiency.
- Include a Service-Oriented Architecture (SOA) with microservices for scalability.
- Support multi-tenancy for different departments and stakeholders.
- Enable data-driven decision-making through visualization and advanced analytics tools.
- Comply with open data standards and ensure future scalability for new smart domains.

3. Technical Requirements

3.1 System Architecture

Provide a layered architecture diagram consisting of:

- Data Acquisition Layer: Collecting data from IoT devices, subsystems, and databases.
- Integration/Middleware Layer: API gateway, message brokers, and ETL engines for data normalization.
- Application Layer: Functional modules for monitoring, analytics, and reporting.
- Visualization Layer: Dashboards, GIS integration, and user interfaces.
- Data Storage Layer: Central data lake and analytics databases.

Architecture must support:

- Cloud-native deployment (private, public, or hybrid).
- Horizontal scalability for future expansion.
- High availability and disaster recovery mechanisms.
- Open APIs for third-party integrations.
- AI and machine learning integration for predictive insights.

3.2 Functional Capabilities

Integration and Data Management:

- Collect and normalize data from multiple smart systems and devices.
- Support real-time data streaming and event-driven operations.
- Enable unified alarm management and incident correlation.

Analytics and Intelligence:

- Provide descriptive, predictive, and prescriptive analytics.
- Implement machine learning models for anomaly detection and demand forecasting.
- Enable performance benchmarking and trend visualization.

Visualization and Dashboards:

- Interactive GIS-based maps for asset and incident tracking.
- 3D digital twin visualization of facilities and zones.
- Configurable dashboards per user role (operator, supervisor, management).

Operational Management:

- Centralized event management system with automated workflows.
- Real-time KPI monitoring for each subsystem.
- Integration with Help Desk (Use Case 20) for service escalation.

APIs and Interoperability:

- Open API gateway for bi-directional communication with third-party systems.
- Support for REST, MQTT, OPC UA, BACnet/IP, and Modbus protocols.
- Adherence to interoperability frameworks (oneM2M, ISO/IEC 30141).

3.3 Integration and Interoperability

The Smart Cities Platform shall integrate with all operational systems, including:

- IoT Platform (Use Case 9)
- C4 Center (Use Case 8)
- BMS (Use Case 19)
- EMS (Use Case 21)
- CAFM (Use Case 2)

- EDMS (Use Case 12)
- ERP (Use Case 15)
- Smart Lighting, Waste, and Irrigation Systems
- Public Help Desk (Use Case 20)
- Digital Payment Platform (Use Case 1)

Integration shall be achieved using:

- Enterprise Service Bus (ESB) or IoT Middleware.
- Standard data formats (JSON, XML, CSV).
- API management tools supporting throttling, authentication, and analytics.

The platform must support northbound integration with external stakeholders and southbound integration with field-level devices.

3.4 Cybersecurity and Data Governance

- Comply with ISO 27001, IEC 62443, and NIST CSF standards.
- Implement Zero Trust Architecture for network and application access.
- Use of TLS 1.3 for all data-in-transit encryption and AES-256 for data-at-rest encryption.
- Role-based access control, MFA, and session monitoring.
- Integration with SIEM systems for security monitoring and log correlation.
- Regular penetration testing, vulnerability assessments, and patching.
- Data governance framework to define ownership, retention, and compliance policies.
- Support for GDPR or equivalent data privacy compliance.

3.5 User Interface and Experience

- Unified, web-based dashboard accessible via browsers and mobile devices.
- Real-time visualization of key KPIs and events.
- Customizable user dashboards based on role and department.
- Multilingual support (Arabic/English).
- Integrated GIS and 3D visualization.
- Notification and alert center with escalation workflows.

4. Operational and Maintenance Requirements

- Define SLAs for system uptime ($\geq 99.95\%$), latency, and performance.
- Include predictive maintenance for hardware and software components.
- Provide full O&M documentation and administrator training.
- Include backup and disaster recovery strategy.
- Offer long-term support and upgrade roadmap.
- Regular system health checks and optimization recommendations.

5. Deliverables

MSI shall provide the following as part of the RFI response:

- End-to-end system architecture diagram (logical and physical).
- Integration framework and API documentation.
- Data model and governance framework.
- Security and compliance documentation.
- Sample dashboards and KPI reports.
- Operations, maintenance, and support plan.
- Reference implementations for similar smart city platforms.

3.15. Use Case 15 – Enterprise Resource Planning (ERP)

1. Background and Objective

The Enterprise Resource Planning (ERP) system is the core business backbone of the Smart Community's operational and administrative framework.

It is designed to centralize and automate all enterprise functions including finance, procurement, human resources, inventory, supply chain, and maintenance ensuring real-time visibility, accuracy, and efficiency in business processes.

The ERP system will also act as the financial and operational integration layer with other digital platforms such as CAFM, IoT, Billing, and Digital Payment Systems.

The objective of this RFI is to gather detailed technical and functional information from the MSI about the proposed ERP solution, covering architecture, modules, integration capabilities, and compliance requirements ensuring alignment with the Smart Community's digital transformation roadmap.

2. Scope of Work

MSI shall design, supply, configure, and commission a comprehensive ERP solution covering, at minimum, the following modules:

- Financial Management: General Ledger, Accounts Payable/Receivable, Fixed Assets, Budgeting.
- Procurement and Inventory Management: Vendor management, purchase requisitions, stock tracking.
- Human Capital Management: Payroll, attendance, leave management, employee performance.
- Asset and Maintenance Management: Integration with CAFM for lifecycle tracking.
- Project and Contract Management: Cost estimation, billing, and contract control.
- Billing and Revenue Management: Integration with Park Operations Billing System (Use Case 10).
- Customer Relationship Management (CRM): Tenant and client engagement tracking.
- Analytics and Reporting: Financial KPIs, dashboards, and audit trails.

The ERP solution must ensure data integrity, interoperability, scalability, and compliance with international accounting and governance standards (e.g., IFRS, ISO 9001).

3. Technical Requirements

3.1 System Architecture

Provide detailed architecture showing all ERP layers (presentation, business logic, data, and integration).

Architecture must support:

- Cloud, on-premises, or hybrid deployment models.
- Multi-entity and multi-currency operations.
- Role-based access control and workflow automation.
- Redundant servers and disaster recovery mechanisms.

- Include an Integration Layer for connectivity with CAFM, IoT, Billing, and Digital Payment systems.
- Must support REST/SOAP APIs for interoperability.
- Include centralized data storage (RDBMS or cloud-based database) with high availability.

3.2 Functional Capabilities

Financial Management:

- Budget creation, forecasting, and variance analysis.
- Real-time consolidation of financial statements.
- Integration with banks and payment gateways.

Procurement and Inventory:

- Automated purchase requests, approvals, and tracking.
- Inventory optimization and vendor evaluation.
- Integration with IoT sensors for real-time asset tracking.

Human Capital Management:

- Employee database, payroll, and HR analytics.
- Time and attendance via biometric or access control integration.

Project and Contract Management:

- Project scheduling, costing, and billing integration.
- Integration with EDMS for document control and contract tracking.

Analytics and Reporting:

- Real-time dashboards with financial KPIs.
- Custom reporting templates and audit logs.
- Predictive insights using AI and machine learning models.

3.3 Integration and Interoperability

The ERP shall integrate seamlessly with:

- Billing System (Use Case 10) – for revenue management.
- CAFM (Use Case 2) – for asset and maintenance cost tracking.
- IoT Platform (Use Case 9) – for automated data input from sensors (e.g., consumption, usage).
- Digital Payment Platform (Use Case 1) – for transaction reconciliation.
- EDMS (Use Case 12) – for document workflows.
- Smart City Platform (Use Case 14) – for analytics and performance dashboards.

Integration must be implemented via open APIs, ETL connectors, or message queues to ensure interoperability.

The system must support data synchronization across all connected platforms in near-real-time.

3.4 Cybersecurity and Data Privacy

- Compliance with ISO 27001, ISO 22301, and GDPR or local data privacy regulations.
- Role-based access control and multi-factor authentication.
- Secure communication using TLS 1.3 and data encryption at rest (AES-256).
- Audit logging and monitoring of all financial transactions.
- Segregation of duties and compliance with financial controls.
- Regular vulnerability scanning, patching, and security audits.
- Disaster recovery and data backup strategy compliant with business continuity requirements.

3.5 User Interface and Experience

- Unified, web-based interface accessible via browsers and mobile devices.
- Role-specific dashboards and workflows (Finance, HR, Procurement, Management).
- Multilingual support (Arabic/English).
- Interactive analytics and KPI widgets.
- Integration with Microsoft Office 365, Power BI, and email systems.
- Intuitive, low-training UI design for end users.

4. Operational and Maintenance Requirements

- Define SLAs for uptime ($\geq 99.9\%$) and response times.
- Provide preventive maintenance schedule for hardware/software.
- Include user training programs and documentation.
- Offer version upgrades, patches, and long-term technical support.
- Define data backup, archiving, and audit procedures.
- Provide support for regulatory audits and financial reporting.
- Establish lifecycle management and performance monitoring.

5. Deliverables

MSI shall include the following as part of the RFI response:

- Full system architecture diagrams (logical and physical).
- Product datasheets and licensing details.
- Integration and API documentation.
- Cybersecurity and compliance certifications.
- O&M and user training plan.
- Example dashboards and reports.
- Reference projects for ERP implementation in smart community or enterprise environments.

3.16. Use Case 16 – Smart Irrigation Systems & Water Management

1. Background and Objective

The Smart Irrigation and Water Management System is a fundamental component of the Smart Community's sustainability and resource optimization strategy. It aims to ensure efficient water consumption, automated irrigation scheduling, and real-time monitoring of water usage across landscapes, parks, and green areas.

This system will leverage IoT sensors, weather data, and analytics to make intelligent irrigation decisions minimizing waste, reducing operational costs, and maintaining optimal greenery conditions throughout the community.

The objective of this RFI is to obtain comprehensive technical and operational information from the MSI regarding the proposed smart irrigation solution, covering architecture, devices, communication protocols, and integration with other smart systems.

2. Scope of Work

MSI shall design, supply, install, and commission a Smart Irrigation and Water Management System that includes:

- Soil moisture sensors, weather stations, and flow meters for data acquisition.
- Automated irrigation controllers capable of dynamic scheduling.
- Integration with the IoT Platform (Use Case 9) and Smart Cities Platform (Use Case 14).
- Remote monitoring and control via web and mobile applications.
- Leak detection and water consumption analytics.
- GIS-based mapping of irrigation zones and assets.
- Water quality monitoring and reporting.
- Compliance with sustainability and green building standards (e.g., LEED, Estidama).

The system shall cover all public landscaped areas, including parks, pathways, and facility surroundings.

3. Technical Requirements

3.1 System Architecture

Provide detailed architecture showing controllers, sensors, communication gateways, and central management software.

Architecture shall support:

- Edge computing for local decision-making.
- IoT-based communication using LoRaWAN, NB-IoT, or MQTT.
- Redundant and fail-safe operation for critical zones.
- Include integration layers for data exchange with IoT Platform and CAFM.
- Support hybrid connectivity (wired/wireless) for different site conditions.

- Include cloud or on-premises hosting options for the central irrigation management application.

3.2 Functional Capabilities

Automated Irrigation Control:

- Intelligent scheduling based on soil moisture, temperature, and humidity.
- Dynamic adjustment according to weather forecasts and rainfall prediction.
- Manual override via web/mobile interface.

Sensor and Data Management:

- Real-time monitoring of soil moisture, pressure, and flow rate.
- Data analytics for water usage and performance optimization.
- Fault detection for leaks or clogged lines.

Water Resource Management:

- Centralized water consumption tracking for each zone.
- Integration with water utility meters and EMS.
- Alerts for abnormal water usage or low tank levels.

GIS and Visualization:

- Integration with GIS maps showing irrigation zones and sensor placement.
- Visualization of live sensor data and irrigation status.
- Historical trend analysis for water usage and system efficiency.

3.3 Integration and Interoperability

The Smart Irrigation System shall integrate with:

- IoT Platform (Use Case 9) – for data collection, device management, and analytics.
- Smart Cities Platform (Use Case 14) – for centralized monitoring and control.
- Energy Management System (Use Case 21) – for power optimization of pumps and controllers.
- CAFM (Use Case 2) – for maintenance scheduling and fault management.
- C4 Center (Use Case 8) – for alarm visibility and incident response.

Integration shall use open protocols and APIs (MQTT, REST, OPC UA).

Data must be normalized and shared through the central IoT middleware to ensure interoperability.

3.4 Cybersecurity and Data Privacy

- Compliance with ISO 27001 and IEC 62443 standards for operational technology (OT) networks.
- Encrypted data communication (TLS 1.3) between devices and central platform.
- Role-based access control and secure device onboarding.
- Regular firmware updates and patching for all IoT devices.

- Secure cloud or on-premises hosting with redundancy.
- Audit trail for all configuration and control actions.

3.5 User Interface and Experience

- Centralized, web-based dashboard for irrigation scheduling and monitoring.
- Interactive GIS interface for zone visualization.
- Real-time alerts for faults, leaks, or weather-based changes.
- Custom reports for water consumption, performance, and sustainability KPIs.
- Multilingual support (Arabic/English).
- Mobile app for field engineers with offline mode capability.

4. Operational and Maintenance Requirements

- Provide maintenance plans for sensors, controllers, and network devices.
- Define SLAs for system uptime ($\geq 99.9\%$) and response times.
- Include procedures for calibration, firmware updates, and data validation.
- Provide user and administrator training programs.
- Submit preventive maintenance and inspection schedules.
- Include spare parts list and warranty coverage.
- Provide remote diagnostics and predictive maintenance tools.

5. Deliverables

MSI shall include the following as part of the RFI response:

- System architecture diagrams and network layouts.
- Product datasheets and technical specifications.
- Integration plan with IoT Platform and EMS.
- Cybersecurity and compliance documentation.
- O&M and training plans.
- Reference projects demonstrating large-scale smart irrigation deployment.

3.17. Use Case 17 – Building Access Control System (BACS)

1. Background and Objective

The Building Access Control System (BACS) is a key component of the Smart Community's physical security framework. It is designed to regulate and monitor access to all facilities, ensuring authorized entry, occupant safety, and real-time security visibility.

The system shall integrate with the centralized Security and Safety Ecosystem, including the C4 Command Center, CCTV, and IoT Platforms, to create a fully interoperable and intelligent access management environment.

The objective of this RFI is to gather detailed technical information from the MSI regarding the proposed Access Control System, including its architecture, security features, and integration methodology ensuring compliance with international standards and alignment with the Smart Community's digital security strategy.

2. Scope of Work

MSI shall design, supply, install, and commission a Building Access Control System (BACS) that covers all public, administrative, and restricted areas within the Smart Community.

The system shall include:

- Access control panels, door controllers, and readers.
- Integration with CCTV, Fire Alarm, and IoT Platforms.
- Centralized management software with real-time monitoring and reporting.
- Support for multiple credential types (RFID, smart card, mobile access, biometrics).
- Alarm management and event correlation with C4 Center.
- High availability architecture with redundancy and failover.
- Compliance with ONVIF, OSDP, and IEC 60839 standards.

3. Technical Requirements

3.1 System Architecture

- Provide detailed architecture showing controllers, readers, and network topology.
- System must support:
 - Distributed intelligence for local decision-making in case of network outage.
 - Centralized server or virtualized platform for configuration and monitoring.
 - Integration via IP-based communication with encryption.
- Include interfaces to integrate with CCTV (for video verification) and Fire Alarm (for emergency door release).
- Support for both online and offline access validation.
- Ensure redundancy for servers and communication links.
- Architecture should allow scalability for future buildings and users.

3.2 Functional Capabilities

Access Management:

- Define access rights by user, group, or schedule.
- Control of doors, gates, turnstiles, and parking barriers.
- Multi-factor authentication options (card + PIN, card + biometric).
- Integration with visitor management and elevator control.

Monitoring and Reporting:

- Real-time event monitoring via centralized GUI.
- Alarm generation for forced door, invalid access, or door held open.
- Comprehensive reporting and audit trail.
- Integration with C4 Center dashboard for situational awareness.

Credential Management:

- Support for RFID, MIFARE DESFire, HID SEOS, and Bluetooth mobile credentials.
- Credential issuance, revocation, and lifecycle management.
- Visitor and contractor access registration.

Emergency and Safety Functions:

- Automatic door release on fire alarm activation.
- Lockdown capability from C4 Center.
- Integration with PAVA (Use Case 13) for voice evacuation messages.

3.3 Integration and Interoperability

The Access Control System shall integrate with:

- C4 Center (Use Case 8): For centralized monitoring and incident correlation.
- CCTV System (Use Case 22): For event-linked video verification.
- Fire Alarm System (Use Case 18): For automatic door release in emergencies.
- IoT Platform (Use Case 9): For event data aggregation and analytics.
- BMS (Use Case 19): For occupancy-based HVAC and lighting control.
- Smart Cities Platform (Use Case 14): For unified security dashboards.

Integration shall utilize open APIs, SDKs, and industry standards (ONVIF, OSDP, BACnet/IP).

Support for event sharing using RESTful APIs or MQTT protocols is required.

3.4 Cybersecurity and Data Privacy

- Compliance with ISO 27001, IEC 62443, and GDPR (or local data privacy laws).
- Secure credential management with AES-256 encryption.
- TLS 1.3 for controller-to-server communication.

- Role-based access control and multi-factor authentication for administrators.
- Full audit logging of configuration and user actions.
- Regular penetration testing, firmware validation, and patch management.
- Segregation of security network (VLANs) from other ICT infrastructure.

3.5 User Interface and Experience

- Web-based management console for operators and administrators.
- Multi-site, multi-building visualization via GIS or floorplan view.
- Multilingual interface (Arabic/English).
- Role-based dashboards (operator, security supervisor, admin).
- Real-time alarm and event notifications via email/SMS/mobile app.
- Integration with C4 video wall for incident visualization.

4. Operational and Maintenance Requirements

- Define SLAs for system uptime ($\geq 99.95\%$) and event response times.
- Provide preventive and corrective maintenance plans.
- Include O&M manuals, as-built documentation, and training.
- Support firmware and software version management.
- Ensure spare parts availability and vendor support for at least 5 years.
- Implement regular system health checks and backup procedures.

5. Deliverables

MSI shall include the following as part of the RFI response:

- System architecture and device layout diagrams.
- Product datasheets for controllers, readers, and software.
- Integration plan with CCTV, C4, and Fire Alarm systems.
- Cybersecurity and compliance documentation.
- Maintenance and training plan.
- Reference projects for large-scale access control implementations.

3.18. Use Case 18 – Addressable Fire Alarm System (ICT Scope)

1. Background and Objective

The Addressable Fire Alarm System (FAS) is a critical life-safety component of the Smart Community infrastructure, responsible for the detection, notification, and coordination of emergency responses across all facilities.

The system must ensure early fire detection, precise localization of incidents, and integration with other safety and operational platforms such as the C4 Command Center, PAVA, BACS, and BMS to enable coordinated emergency management and evacuation procedures.

The objective of this RFI is to collect detailed technical information from the MSI regarding the proposed fire alarm system architecture, compliance standards, integration capabilities, and operational reliability, ensuring alignment with international life-safety regulations and smart community integration requirements.

2. Scope of Work

MSI shall design, supply, install, and commission a fully addressable Fire Alarm System covering all buildings and public spaces across the Smart Community.

The system shall:

- Detect fire, smoke, heat, and gas using intelligent sensors.
- Provide precise zone and device-level identification.
- Support automatic activation of PAVA for evacuation messages.
- Interface with the C4 Center and BMS for event monitoring and coordination.
- Comply with international codes and standards (NFPA 72, EN 54, UL 864).
- Include redundant loops, power supplies, and network communication.
- Support integration with suppression systems where applicable.

3. Technical Requirements

3.1 System Architecture

Provide architecture showing panels, detectors, loops, power supplies, and network topology.

System must be fully addressable and support:

- Loop-based detection with isolation modules.
- Automatic device addressing and mapping.
- Peer-to-peer communication between panels.
- IP-based networking between buildings and C4 Center.
- Include redundancy for main and sub-panels with failover capabilities.
- System should allow remote configuration and monitoring.
- Design must comply with local civil defense authority regulations.

3.2 Functional Capabilities

Detection and Alarm:

- Smoke, heat, flame, and gas detection sensors.
- Sounders, beacons, and manual call points.
- Sensitivity adjustment based on environmental conditions.

Event Management:

- Real-time alarm, fault, and pre-alarm notification.
- Automatic triggering of PAVA (Use Case 13) for evacuation.
- Relay outputs to control HVAC shutdown and door release.

Monitoring and Control:

- Centralized monitoring through the C4 Center (Use Case 8).
- Graphical interface showing floor plans and device locations.
- Historical log of events and system health.

Integration Capabilities:

- Integration with BMS (Use Case 19) for HVAC and smoke control.
- Integration with BACS (Use Case 17) for emergency egress control.
- Integration with IoT Platform (Use Case 9) for data analytics.

3.3 Integration and Interoperability

The Fire Alarm System shall integrate with:

- C4 Center (Use Case 8): For centralized visualization, alarm correlation, and incident reporting.
- PAVA (Use Case 13): For automatic voice announcements and evacuation instructions.
- BMS (Use Case 19): For smoke management and HVAC control.
- BACS (Use Case 17): For door release and lockdown coordination.
- Smart Cities Platform (Use Case 14): For global monitoring and analytics.

Integration shall be achieved through open protocols (Modbus TCP/IP, BACnet/IP, REST API).

All network interfaces must be isolated within the ICT infrastructure with VLAN segregation.

3.4 Cybersecurity and System Reliability

- Compliance with NFPA 72, EN 54, UL, and ISO 27001.
- Redundant network and power configurations.
- Use of secure communication protocols for IP-based links.
- Role-based access control for configuration and maintenance.
- System hardening and firmware signing to prevent tampering.
- Secure remote access (VPN or dedicated link) for authorized personnel only.

- Periodic testing, calibration, and certification as per local authority requirements.

3.5 User Interface and Experience

- Graphical Fire Alarm Control and Monitoring (FCMS) interface.
- Floor plan visualization with real-time device status (alarm/fault/normal).
- Multilingual interface (Arabic/English).
- Event prioritization and alarm acknowledgment workflow.
- Integration with C4 video wall for situational awareness.
- Mobile alerting for facility managers and emergency responders.

4. Operational and Maintenance Requirements

- Define SLAs for system availability ($\geq 99.99\%$) and fault response time.
- Provide preventive maintenance plan including device cleaning and testing.
- Include system calibration and re-commissioning schedules.
- Provide O&M documentation, as-built drawings, and training.
- Include spare parts, warranty, and technical support for a minimum of 5 years.
- Implement continuous monitoring for system health and device connectivity.

5. Deliverables

MSI shall include the following as part of the RFI response:

- Complete system architecture and block diagrams.
- Product datasheets for panels, detectors, and interfaces.
- Integration plan with PAVA, C4, and BMS.
- Compliance certificates (NFPA, EN 54, UL).
- Cybersecurity and redundancy design documentation.
- Maintenance and training plan.
- Reference projects for similar addressable fire alarm implementations.

3.19. Use Case 19 – Building Management System (BMS)

1. Background and Objective

The Building Management System (BMS) is the central operational control framework for monitoring and managing all electromechanical and environmental systems within the Smart Community's buildings.

Its purpose is to enhance energy efficiency, comfort, safety, and asset lifecycle performance through automation, centralized control, and real-time analytics.

The system will integrate with other platforms such as IoT, EMS, C4 Center, and Smart Cities Platform to achieve a unified smart infrastructure ecosystem.

The objective of this RFI is to gather detailed information from the MSI regarding the proposed BMS solution covering architecture, functionalities, interoperability, and cybersecurity ensuring alignment with smart city and sustainability objectives.

2. Scope of Work

MSI shall design, supply, install, and commission a comprehensive Building Management System (BMS) for all buildings within the Smart Community.

The system shall include:

- Monitoring and control of HVAC, lighting, elevators, pumps, and electrical systems.
- Integration with Fire Alarm, Access Control, and Energy Management Systems.
- Real-time data acquisition from field devices (sensors, controllers, meters).
- Trend logging, analytics, and reporting.
- Alarm management and event prioritization.
- Support for local and centralized control via the C4 Center.
- Compliance with ASHRAE, BACnet, and ISO 16484 standards.

3. Technical Requirements

3.1 System Architecture

Provide detailed system architecture including controllers, sensors, communication buses, and supervisory servers.

Architecture must support:

- Hierarchical control (field, automation, and management layers).
- Distributed controllers with local intelligence.
- IP-based communication with redundant backbone.
- Include open communication protocols (BACnet/IP, Modbus TCP, OPC UA).
- Support virtualization and cloud connectivity for data storage and analytics.
- Ensure scalability for additional subsystems and future buildings.
- Integrate securely with the Smart Cities Platform via API or middleware.

3.2 Functional Capabilities

Monitoring and Control:

- HVAC, chillers, air handling units, and ventilation.
- Lighting systems with occupancy-based automation.
- Pumps, water tanks, and elevators.
- Generators, UPS, and power monitoring.

Alarms and Events:

- Automatic detection of faults and abnormal conditions.
- Alarm classification, prioritization, and notifications.
- Integration with C4 Center for centralized visualization.

Energy Optimization:

- Load scheduling and peak demand control.
- Integration with EMS for energy analytics and benchmarking.
- Predictive maintenance based on equipment runtime and performance.

Analytics and Reporting:

- Historical trend analysis and data visualization.
- KPI dashboards for energy, comfort, and system performance.
- Integration with BI tools for data-driven insights.

3.3 Integration and Interoperability

The BMS shall integrate with:

- Fire Alarm System (Use Case 18) – for HVAC shutdown and smoke control.
- Access Control System (Use Case 17) – for occupancy-based automation.
- IoT Platform (Use Case 9) – for device-level data exchange.
- Energy Management System (Use Case 21) – for energy analytics and load optimization.
- C4 Center (Use Case 8) – for centralized alarm and control visualization.
- Smart Cities Platform (Use Case 14) – for unified command and monitoring.

Integration shall utilize open standards and APIs (BACnet/IP, OPC UA, REST).

Data must be normalized within the IoT or middleware layer for interoperability.

3.4 Cybersecurity and Data Privacy

- Compliance with ISO 27001, IEC 62443, and NIST Cybersecurity Framework.
- Segregated control network with firewalls and VLANs.
- LS 1.3 encryption for data transmission.
- Role-based access control for users and administrators.

- Secure firmware and controller hardening.
- Regular security patching, auditing, and vulnerability testing.
- Backup and disaster recovery procedures for configuration data.

3.5 User Interface and Experience

- Web-based graphical user interface with floorplan visualization.
- Real-time status and alarm dashboards.
- Multilingual support (Arabic/English).
- Role-based views for operators, maintenance, and management.
- Integration with video wall and mobile devices for remote access.
- KPI and performance visualization with drill-down capabilities.

4. Operational and Maintenance Requirements

- Define SLAs for uptime ($\geq 99.9\%$) and control latency.
- Provide preventive and corrective maintenance schedules.
- Include O&M documentation and as-built drawings.
- Provide user training and certification programs.
- Include firmware update, calibration, and spare parts plan.
- Implement predictive maintenance analytics.
- Provide warranty and support for a minimum of 5 years.

5. Deliverables

MSI shall include the following as part of the RFI response:

- System architecture and network topology diagrams.
- Product datasheets and technical specifications.
- Integration plan with IoT, EMS, and C4 systems.
- Cybersecurity compliance documents.
- Operations and maintenance plan.
- Example dashboards and reports.
- Reference projects demonstrating large-scale BMS implementations.

3.20. Use Case 20 – Public Help Desk & Call Centre System Solution

1. Background and Objective

The Public Help Desk and Call Centre System is a critical component of the Smart Community's customer experience and operational support framework.

It provides a centralized communication and service platform for residents, visitors, and tenants to report issues, request services, and obtain information efficiently.

The system shall act as the primary interface between the community and its operational departments, integrating with other smart systems such as CAFM, IoT, ERP, and C4 Center to enable automated service requests, real-time status tracking, and data-driven performance analysis.

The objective of this RFI is to obtain detailed technical, functional, and operational information from the MSI regarding the proposed Help Desk and Call Centre solution — ensuring it supports omnichannel communication, service automation, and integration with the Smart Community's unified digital ecosystem.

2. Scope of Work

MSI shall design, supply, and implement a Public Help Desk and Call Centre Solution that provides:

- Centralized platform for managing all customer interactions (calls, emails, mobile app, web, chatbot).
- Integrated ticketing and case management system with workflow automation.
- Integration with CAFM for maintenance and service requests.
- Integration with C4 Center for incident and emergency reporting.
- Integration with ERP and Billing Systems for account and payment inquiries.
- Service performance analytics and reporting dashboards.
- Multilingual (Arabic/English) interface and IVR support.

3. Technical Requirements

3.1 System Architecture

Provide detailed architecture showing communication channels, ticketing engine, databases, and integration layers.

System must support:

- Omnichannel communication (voice, email, SMS, mobile app, chatbot, social media).
- Centralized CRM/ticketing engine for unified case management.
- Cloud or hybrid deployment for scalability and resilience.
- High-availability design with redundant communication servers.
- Support integration with telephony systems (SIP/VoIP).
- Include workflow automation and service routing based on issue type or location.

- API integration with Smart Cities Platform (Use Case 14) for unified monitoring.

3.2 Functional Capabilities

Call Centre Operations:

- IVR-based call routing and self-service options.
- Call recording, logging, and reporting.
- Integration with mobile app and web portal for ticket creation and tracking.

Help Desk and Ticketing:

- Centralized service request management.
- Automated ticket assignment based on service type and priority.
- Integration with CAFM for facility and maintenance requests.
- SLA tracking, escalation, and resolution reporting.

CRM and Customer Interaction:

- Centralized customer database and history tracking.
- Integration with ERP for billing or account inquiries.
- Automated communication (notifications, acknowledgments).

Analytics and Reporting:

- Real-time dashboards showing call volumes, response times, and SLA performance.
- Trend analysis and customer satisfaction metrics.
- AI-driven sentiment analysis for feedback evaluation.

3.3 Integration and Interoperability

The Help Desk and Call Centre System shall integrate with:

- CAFM (Use Case 2): For automatic work order generation and service tracking.
- ERP (Use Case 15): For financial and billing-related requests.
- IoT Platform (Use Case 9): For automatic alert-based service tickets.
- C4 Center (Use Case 8): For incident management and emergency escalation.
- Smart Cities Platform (Use Case 14): For unified visualization and reporting.
- Mobile App (Use Case 5): For resident and visitor access to support channels.

Integration shall utilize open APIs (REST, SOAP) and message queues for data synchronization.

All systems must support event-driven architecture to allow real-time interaction.

3.4 Cybersecurity and Data Privacy

- Comply with ISO 27001, GDPR, and PCI DSS where applicable.
- Encrypt all communications (TLS 1.3 for web, SRTP for voice).

- Implement secure authentication and role-based access control.
- Maintain audit trails for all customer interactions.
- Secure customer data storage and anonymization policies.
- Periodic penetration testing and patch management.
- Backup and disaster recovery in accordance with ITIL best practices.

3.5 User Interface and Experience

- Unified web-based dashboard for operators and supervisors.
- Multi-channel inbox for calls, messages, and chat requests.
- Configurable workflows and SLA monitoring.
- Multilingual (Arabic/English) user interface.
- Self-service customer portal and mobile app integration.
- Visual dashboards for ticket status, KPIs, and team performance.

4. Operational and Maintenance Requirements

- Provide SLAs, response time and warranty for at least 5 years from the date of handover.
- Include preventive and corrective maintenance schedules.
- Offer operator and administrator training programs.
- Provide documentation for system configuration, workflow design, and troubleshooting.
- Include version upgrade, patch management, and long-term support.
- Define disaster recovery and continuity procedures.
- Implement quality assurance and performance review mechanisms.

5. Deliverables

MSI shall include the following as part of the RFI response:

- System architecture and integration diagrams.
- Product datasheets and software specifications.
- Sample reports and dashboard templates.
- Cybersecurity and compliance documentation.
- Operations and maintenance plan.
- Training and support program.
- Reference projects for public call centre and help desk deployments.

3.21. Use Case 21 – Energy Management System (EMS)

1. Background and Objective

The Energy Management System (EMS) is a core element of the Smart Community's sustainability and operational efficiency framework.

It provides real-time monitoring, analysis, and optimization of energy consumption across all facilities, enabling informed decision-making, cost savings, and environmental compliance.

The EMS shall collect and analyze energy data from meters, IoT devices, and building systems, integrating with BMS, IoT Platform, and the Smart Cities Platform to enable advanced analytics, demand response, and predictive optimization.

The objective of this RFI is to obtain comprehensive details from the MSI regarding the proposed EMS architecture, data collection methods, interoperability, and performance analytics — ensuring alignment with smart city sustainability and carbon reduction goals.

2. Scope of Work

MSI shall design, supply, install, and commission a comprehensive Energy Management System (EMS) that enables:

- Real-time energy consumption monitoring and control.
- Integration with building-level metering and sub-metering systems.
- Data aggregation from BMS, IoT, and renewable energy sources.
- Energy analytics, reporting, and KPI visualization.
- Load forecasting and peak demand management.
- Alarm management for abnormal consumption patterns.
- Integration with Smart Cities Platform for consolidated reporting.
- Compliance with ISO 50001 (Energy Management Systems).

The system shall cover all community facilities, buildings, and infrastructure utilities.

3. Technical Requirements

3.1 System Architecture

Provide detailed architecture showing meters, gateways, data loggers, servers, and application layers.

System shall support:

- Hierarchical data collection (building → zone → meter).
- Real-time and historical data storage for analytics.
- Cloud or on-premises deployment with scalability for expansion.
- Integration layer for interoperability with other platforms (IoT, BMS, ERP).
- Include redundancy, failover, and disaster recovery mechanisms.
- Support open standards and protocols: Modbus TCP/IP, BACnet/IP, MQTT, OPC UA.

3.2 Functional Capabilities

Monitoring and Measurement:

- Real-time energy monitoring for electrical, water, and gas utilities.
- Sub-metering and tenant-level consumption tracking.
- Power quality monitoring (voltage, current, harmonics).

Analytics and Optimization:

- Load profiling, demand forecasting, and efficiency analysis.
- Identification of energy wastage and performance anomalies.
- Benchmarking and KPI-based performance evaluation.
- Integration with AI/ML tools for predictive energy optimization.

Reporting and Visualization:

- Dashboards for energy KPIs (consumption, efficiency, cost, carbon).
- Comparison between facilities, time periods, and usage categories.
- Automated report generation (daily, monthly, annual).

Demand Management:

- Load shedding and control strategies.
- Integration with renewable energy sources and energy storage.
- Alerting and automated actions based on threshold violations.

3.3 Integration and Interoperability

The EMS shall integrate with:

- BMS (Use Case 19): For data synchronization and equipment-level control.
- IoT Platform (Use Case 9): For device connectivity and analytics.
- Smart Cities Platform (Use Case 14): For consolidated KPI dashboards and sustainability reporting.
- ERP (Use Case 15): For energy cost allocation and billing.
- C4 Center (Use Case 8): For monitoring and alarm visibility.

Integration must be achieved via open APIs, middleware, or message brokers.

Data exchange should support real-time synchronization and secure transfer over IP networks.

3.4 Cybersecurity and Data Privacy

- Comply with ISO 27001, IEC 62443, and NIST CSF frameworks.
- Secure data communication using TLS 1.3 and AES-256 encryption.
- Role-based access control and multi-factor authentication.
- Audit trails for all configuration and control actions.
- Segregated OT and IT network zones.

- Regular vulnerability assessments and patch management.
- Backup and disaster recovery as per business continuity standards.

3.5 User Interface and Experience

- Web-based, centralized dashboard with multilingual support (Arabic/English).
- Visualization of real-time and historical data in graphs, maps, and tables.
- Role-based dashboards for energy managers, operators, and executives.
- Customizable KPI widgets and alerts.
- Mobile application support for remote monitoring.
- Integration with C4 video wall and Smart Cities visualization layer.

4. Operational and Maintenance Requirements

- Define SLAs for uptime ($\geq 99.9\%$) and data accuracy.
- Include preventive and corrective maintenance plans.
- Provide system documentation, training, and user manuals.
- Establish calibration and verification schedules for meters and sensors.
- Implement automated health checks and performance reports.
- Include warranty, updates, and long-term technical support (≥ 5 years).
- Conduct periodic energy audits and performance reviews.

5. Deliverables

MSI shall include the following as part of the RFI response:

- System architecture and network diagrams.
- Product datasheets for meters, gateways, and software components.
- Integration plan with BMS, IoT, and Smart City Platforms.
- Compliance certificates (ISO 50001, IEC, UL).
- O&M and training plans.
- Example dashboards and analytical reports.
- Reference projects for EMS deployment in large-scale communities.

3.22. Use Case 22 – Security Video System & Control System (CCTV Surveillance – ICT Scope)

1. Background and Objective

The Security Video System (CCTV Surveillance) is one of the most critical components in the Smart Community's integrated safety and security infrastructure.

It provides continuous video monitoring, recording, analytics, and situational awareness to ensure public safety, asset protection, and incident response coordination.

The system will serve as the primary visual intelligence layer integrated with the C4 Command Center (Use Case 8) and other safety subsystems such as Access Control (Use Case 17), Fire Alarm (Use Case 18), and IoT Platform (Use Case 9).

The objective of this RFI is to obtain comprehensive information from the MSI regarding the proposed CCTV system architecture, functional capabilities, analytics features, integration strategy, and compliance with cybersecurity and data privacy regulations.

2. Scope of Work

MSI shall design, supply, install, and commission a complete CCTV Surveillance System covering all public, commercial, and operational zones of the Smart Community.

The system shall include:

- Fixed, PTZ, multi-sensor, and panoramic IP cameras.
- Network Video Recorders (NVRs) or centralized Video Management System (VMS).
- Video analytics and AI-based event detection.
- Integration with Access Control, Fire Alarm, and C4 Center.
- Secure network infrastructure for video transport.
- Compliance with ONVIF and local security authority standards.

3. Technical Requirements

3.1 System Architecture

Provide detailed architecture showing cameras, NVRs, VMS servers, storage, and network topology.

System must support:

- Fully IP-based architecture with PoE and redundant network paths.
- Hierarchical design: edge (camera) → aggregation (NVR/VMS) → core (C4 Center).
- Centralized Video Management System (VMS) for control, playback, and analytics.
- Scalable storage infrastructure (NAS/SAN) with RAID and redundancy.
- High-availability failover for VMS and recording servers.
- Video transmission shall use secure protocols (HTTPS, SRTP) and bandwidth optimization (H.265, Smart Codec).

- Support for hybrid connectivity (wired/wireless) for remote or outdoor locations.

3.2 Functional Capabilities

Core Surveillance Features:

- Live monitoring, playback, and multi-site management.
- Continuous, event-based, and motion-triggered recording.
- Multi-view display (map, grid, timeline).
- Video Analytics and AI:
 - Intrusion detection, loitering, line crossing, and object removal.
 - Face detection and recognition (where permitted by local law).
 - License plate recognition (LPR/ANPR).
 - People counting, heat mapping, and crowd density analysis.
 - Vehicle and object classification.

Event Management:

- Automated event generation and alarm escalation.
- Integration with C4 for correlated incident response.
- Custom workflows and notifications.

Recording and Storage:

- Support for continuous 24/7 recording.
- Adjustable frame rates and retention policies (minimum 90 days).
- RAID-5/6 storage with hot-swappable drives.
- Redundant power and network interfaces.

3.3 Integration and Interoperability

The CCTV System shall integrate with:

- C4 Command Center (Use Case 8): For centralized visualization, event correlation, and video wall display.
- Access Control (Use Case 17): For visual verification of access events.
- Fire Alarm (Use Case 18): For live video pop-up upon fire detection.
- IoT Platform (Use Case 9): For analytics and event sharing.
- Smart Cities Platform (Use Case 14): For KPI visualization and situational dashboards.
- Public Help Desk (Use Case 20): For linking incidents to video evidence.

Integration must use ONVIF Profile S/G/T, REST APIs, and SDKs provided by the VMS vendor.

All video metadata and events shall be accessible through standardized APIs for analytics and AI integration.

3.4 Cybersecurity and Data Privacy

- Comply with ISO 27001, NIST, and IEC 62443 for OT cybersecurity.
- Encrypt video streams and stored data using AES-256 and TLS 1.3.
- Implement role-based access control, strong password policies, and multi-factor authentication.
- Maintain audit logs for all user actions and video exports.
- Ensure secure onboarding of cameras with signed certificates.
- Regular vulnerability scanning, firmware updates, and patch management.
- Comply with local data protection laws for video retention and export handling.

3.5 User Interface and Experience

- Intuitive, web-based VMS interface for operators and supervisors.
- Multilingual (Arabic/English) support.
- Customizable layouts for live and playback views.
- GIS-based map integration showing camera locations and alarm status.
- Integration with C4 Center video wall for situational monitoring.
- Real-time notifications and event-based video pop-ups.
- Centralized dashboard for system health, storage, and performance monitoring.

4. Operational and Maintenance Requirements

- Define SLAs for uptime ($\geq 99.9\%$) and video retention.
- Include preventive and corrective maintenance schedules.
- Provide O&M manuals, as-built drawings, and training programs.
- Implement system health monitoring and automated alerts.
- Provide spare parts, warranty, and vendor support for ≥ 5 years.
- Regular system audits for firmware, configuration, and performance.
- Include scalability plan for future camera or site expansion.

5. Deliverables

MSI shall include the following as part of the RFI response:

- Complete system architecture and network diagrams.
- Product datasheets for cameras, NVRs, and VMS components.
- Integration plan with C4, Access Control, and IoT systems.
- Cybersecurity and compliance documentation.
- Operations and maintenance plan.
- Example dashboards, layouts, and analytics reports.
- Reference projects demonstrating large-scale IP surveillance deployment.